Neural Codes of Uncertainty in Visual Perception

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Technical Abstract

The neural representation of uncertainty constitutes one of the fundamental computational problems in neuroscience. Whilst extensive behavioural evidence demonstrates that biological neural systems perform near-optimal probabilistic inference under uncertainty, the underlying implementation remains contested. This thesis develops a computational framework grounded in Bayesian principles that forms part of an analysis pipeline to help researchers investigate a central question: how does the primary visual cortex (V1) encode uncertainty about sensory stimuli - through static population activity patterns at single time points (spatial coding) or through dynamic stochastic sampling over time (temporal coding)? We develop a computational analysis framework grounded in Bayesian principles, that would provide a robust methodology for adjudicating between these competing hypotheses. The framework developed here serves to further the experimental research of Koblinger et al. (2024), who designed an orientation discrimination task in which mice categorise visual gratings as "left" or "right" based on a vertical decision boundary (for a reward). For real biological data, which is measured from the mice during the task, the underlying ground truth encoding processes are not known; therefore a key contribution of this research is to provide computational encoding and decoding frameworks using synthetic data (where we generate the data according to known ground truth encoding schemes), which will provide an analysis pipeline that can be used on real biological data. This work extends the Gaussian Scale Mixture (GSM) model proposed by Orbán et al. (2016), which assumes that visual images $\mathbf{x}$ can be represented as a linear combination of basis functions $\mathbf{y}$, multiplied by contrast (gain) and Gaussian additive noise. The first essential innovation lies in the introduction of orientation-conditioned priors $p(\mathbf{y}|\theta)$ over latent neural features, a modification that proves mathematically necessary for computing ground-truth posterior distributions over stimulus orientation. This gave the original unconditional prior $p(\mathbf{y}) \sim \mathcal{N}(\mathbf{0}, \mathbf{C})$ the necessary structure (mean and covariance become dependent on stimulus orientation), which subsequently allowed for the computation of $p(\theta|\mathbf{x})$ through Bayesian inversion, using both closed form and numerical methods. Looking at properties of our orientation conditioned prior in latent space confirmed that this parametrisation had the necessary construction.

Following the extension of the ideal observer model to orientation θ, we incorporated categorical decision making, reflecting the need to analyse uncertainty representations when we ultimately map continuous sensory information onto discrete behavioural responses (the mice have to perform the left/right mapping based on the grating orientation). A hierarchical generative model is developed where a binary decision variable $d \in \{-1, +1\}$ constrains the orientation distribution, which in turn determines latent features through orientation-conditioned priors. And then together with contrast gain, the observed image $\mathbf{x}$ is generated through the GSM process. This hierarchical structure enables the computation of both perceptual uncertainty (over orientation as before) and decision uncertainty (binary posterior over choice $p(d|\mathbf{x})$), providing ground-truth supervision signals to train neural decoders.

After validating these models, a synthetic neural data generator was implemented for the temporal coding scheme. Under temporal coding, each latent activation represents an independent sample from the posterior distribution, $p(\mathbf{y}|\mathbf{x})$, with each sample potentially representing neural activity at successive time points. The number of samples is constrained to three, reflecting the effective sample size limitation imposed by calcium imaging temporal resolution in the experimental paradigm of Koblinger et al. (2024).

With a working temporal neural data generator, we explored decoding strategies using neural networks as function approximators (*neural decoders*). In the first instance we aimed to recover the perceptual posterior $p(\theta|\mathbf{x})$ using samples from $p(\mathbf{y}|\mathbf{x})$, and then we also trained networks to recover the decision posterior $p(d|\mathbf{x})$, using the full extended ideal observer model for the supervision signal. Whilst our ground truth encoding remains temporal, two decoding strategies were used. To test the temporal decoding, we trained a neural decoder to learn $p(\theta|\mathbf{x})$ (or $p(d|\mathbf{x})$) when given three samples of $\mathbf{y}$ from $p(\mathbf{y}|\mathbf{x})$. The network makes three predictions of the relevant posterior, and we then aggregate these at the output of the networks in probability space. To test spatial decoding assumptions, the same temporally-encoded samples are averaged to create instantaneous population vectors, and these are inputted into a separately trained neural decoder.

The findings strongly support temporal decoding for perceptual uncertainty representation. Across various settings of orientation, contrast and aperture, temporal decoders achieved mean normalised Kullback-Leibler (KL) divergences of 0.103 ± 0.094 (with respect to the ground truth), significantly outperforming spatial decoders at 0.128 ± 0.165 ($p < 0.001$, paired permutation test with 5,000 permutations). This provided independent computational validation of Koblinger et al.'s experimental findings in mouse V1. Further exploration targeted the decomposition of the perceptual temporal decoder through an ablation study, and it revealed the importance of sample integration. This study compared two trained decoders - one on single samples of $\mathbf{y}$ and a second that uses three temporal samples as per the experimental set up. While both schemes accurately recovered posterior means (correlation with ground truth: three-sample r=0.983, single-sample r=0.938), the three-sample decoder dramatically outperformed the single-sample decoder in variance estimation (r=0.928 vs. 0.736). This favours the view that neural variability carries essential uncertainty information rather than representing mere noise, lending support to the temporal coding scheme.

Compared to our results in the perceptual decoding, an unexpected reversal emerged for decision uncertainty: spatial decoding consistently outperformed temporal decoding when decoding binary categorical posteriors (achieving lower KL divergences). The superiority of spatial decoding for decision uncertainty - despite temporal encoding - suggests that averaging may help for coarse categorical information. This raises important questions about the generality of neural networks as function approximators for neural decoding, and provides a point that needs to be explored and validated in future work to verify whether the result was an artifact of our methods.

The research has focused on spatial versus temporal decoding of temporally-encoded data. To investigate true spatial encoding, a preliminary Distributed Distributional Code (DDC) framework was implemented. DDCs encode posterior distributions through high-dimensional non-linear projections without any temporal sampling. While spatial decoders so far processed averaged temporal samples, DDCs would generate genuinely spatial codes where single population vectors fully represent uncertainty. Future work training decoders on DDC representations would provide a fully testable computational framework.

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Nomenclature

Abbreviations

BIO	Bayesian Ideal Observer
DDC	Distributed Distributional Code
ESS	Effective Sample Size
GSM	Gaussian Scale Mixture
KL	Kullback-Leibler (divergence)
MLP	Multi-Layer Perceptron
PPC	Probabilistic Population Code
V1	Primary Visual Cortex

Symbols

$\mathbf{A}$	$D_x \times D_y$ matrix of basis functions (Gabor filters)
$\mathbf{C}$	Covariance matrix of latent features
C_θ	Orientation-conditioned covariance matrix
d	Decision variable, $d \in \{-1, +1\}$
D_x	Dimensionality of image space
D_y	Dimensionality of latent feature space
D_z	Number of discretisation points for contrast
f_{NN}	Neural network decoder function
K	Number of encoding functions in DDC
N	Number of samples
n_d	Number of orientations in decision category d
$p(\cdot)$	Probability distribution
$r^{(i)}$	DDC encoding function output
$T^{(i)}$	DDC nonlinear encoding function
$\mathbf{w}^{(i)}$	Random projection vector for DDC
$\mathbf{x}$	Observed image vector
$\mathbf{y}$	Latent feature activation vector
$\bar{\mathbf{y}}$	Averaged latent features
z	Contrast scaling variable
$\boldsymbol{\epsilon}$	Additive Gaussian noise
θ	Stimulus orientation, $\theta \in \{1°, ..., 180°\}$
Θ_d	Set of orientations consistent with decision d
λ	Wavelength of grating
$\boldsymbol{\mu}_\theta$	Orientation-conditioned mean
$\boldsymbol{\mu}_z$	Posterior mean conditioned on contrast

1 Introduction

1.1 Motivation

Perception can be defined as a process of inference under uncertainty. In particular we consider perception in the V1 visual cortex, which is the primary processor of visual information in the brain and is responsible for processing basic features such as edges and lines. When presented with stimuli, the input to V1 is often corrupted by noise along with various other aspects of the environment (called nuisance variables). These nuisance variables introduce uncertainty under which the brain must infer the true state of the world. For example, factors such as the contrast of an image or the portion of the scene visible (aperture) can affect the magnitude of perceptual uncertainty, and thus impacts the reliability of the sensory information. If neurons did not signal uncertainty, one would expect perception to break down in noisy environments; but instead when presented with visual stimuli our brain weights cues in a near statistically optimal way [1]. Extensive research in neuroscience and psychology have shown that animal brains perform this weighting in a way that is consistent with a Bayesian inference framework [2][3]. Termed the *Bayesian brain hypothesis*, it is proposed that neural circuits implement probabilistic inference, combining their internal model with sensory evidence to compute probabilistic posterior estimates of the states of the world.

Whilst this computational process of Bayesian inference is consistent with much of the contemporary research on quantification of uncertainty in perception and decision, an open question remains on exactly *how this posterior distribution is encoded in biological neural activity?*.

In the following sections, we use the Bayesian brain foundation to examine how uncertainty might be encoded in biological neural circuits. We start by building on the Orbán et al. 2016 [4] Bayesian model of neural computation and evidence for Bayesian inference in the brain. We then distinguish between uncertainty about perceptual quantities versus uncertainty about decisions, an important conceptual difference for neural coding. After that, we review proposed coding schemes for uncertainty. Two main coding schemes have been proposed regarding the neural representation of this uncertainty information. The **spatial coding scheme** posits that uncertainty is represented at a single time point across a population of neurons[3][5] that encode full probability distributions over the stimulus. Conversely, the **temporal coding scheme** takes the view that this probabilistic uncertainty is represented through sampling based codes - neural responses individually are samples from a posterior distribution over the stimulus [4][6] and at any one moment, this activity provides a noisy snapshot on the stimulus rather than encoding a full posterior (as per the spatial scheme).

1.2 Research Objectives and Contributions

Grounded in Bayesian optimality, both the temporal and spatial coding schemes could in principle produce similar behavioural results despite the different internal neural

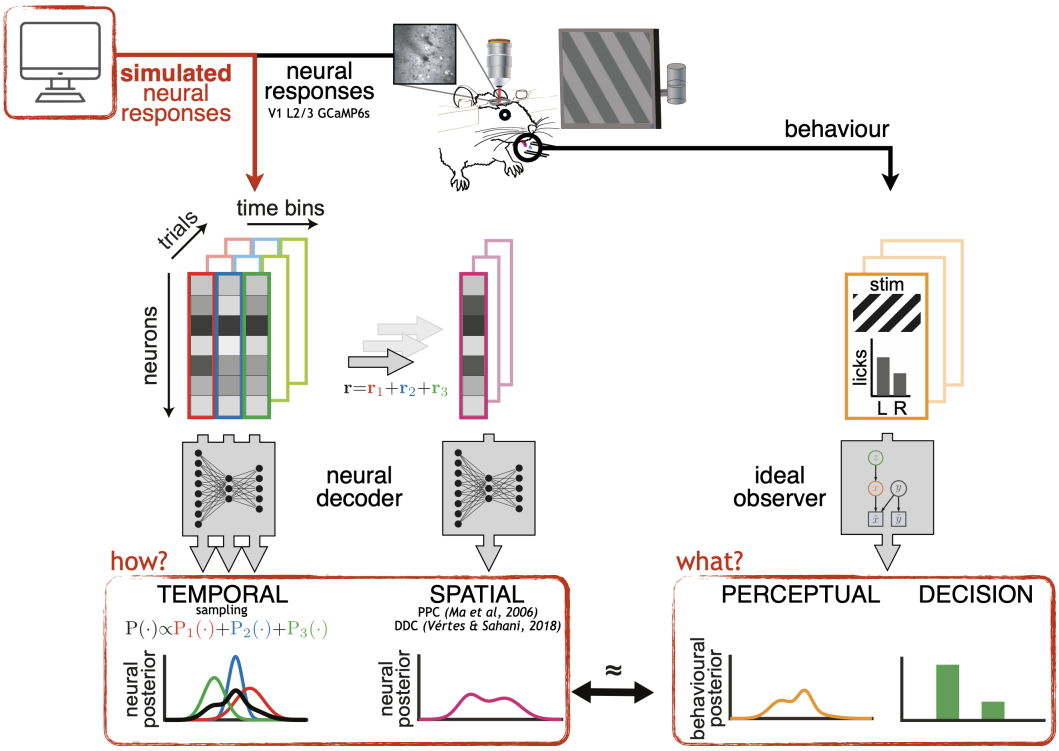


Figure 1: Overview of the behavioural experiment pipeline proposed by Koblinger et al.[10].

computations that may be taking place. In biological systems, it is a difficult task to directly disentangle these two coding paradigms, so to distinguish whether V1 encodes uncertainty temporally or spatially, one needs an experimental and computational methodology that is able to test each hypothesis in a data driven manner under controlled conditions.

The experimental approach has been proposed by Koblinger 2024[10], who designed an orientation estimation based task. This behavioural study is illustrated in Figure 1, where trained mice are shown grating stimuli of systematically varying contrast and aperture (nuisance variables) and they are rewarded when they correctly report the grating direction (left or right based on a vertical decision boundary). By analysing the neural data of the animal using calcium imaging techniques (GCaMP6s) in conjunction with monitoring the behaviour (e.g. decision variability) as a function of the manipulations in the nuisance variables, downstream examination can allow one to recover how uncertainty in both perception and decision is represented. However, direct interpretation of biological data in search for the ground truth encoding scheme is not a tractable task and we need robust analysis pipelines to help us analyse how each coding scheme would manifest in neural responses.

In this project we adopt a modelling and computational approach to the problem. As shown on the right hand side of Figure 1 we construct a *Bayesian ideal observer* (BIO) model for the task, which will be used to extract a ground truth posterior distribution initially over the orientation of a given visual stimulus (perceptual uncertainty) and we extend this model to also have a binary posterior over the decision variable. We proceed to posit the two different coding mechanisms (as shown in Figure 2): the temporal code and spatial code. From these coding hypotheses, we generate synthetic neural data (the methodology and development of which is a key objective of this project) as seen on

the left hand side of Figure 1. These generators produce realistic synthetic data, and importantly we know the coding hypothesis underpinning their construction.

Neural decoders (in this case feed-forward neural networks) are trained to decode the perceptual (or decision) uncertainty from the synthetic neural activity. As shown in Figure 1, in the temporal case we decode three samples and aggregate their posteriors to formulate the final posterior, and in the spatial case we adopt the configuration that all uncertainty information is encoded at one time point, hence we decode the posterior from one averaged sample as shown. Finally, by comparing decoder performance and assessing each decoder's ability to recover the ground truth distributions over orientation and decision (from latent activation vectors $\mathbf{y}$), we can assess the validity of our analysis pipeline. These trained networks can then be used on biological data where we do not know the true coding mechanism, and we can assess which decoder exhibits superior performance in recovering uncertainty.

Key Contributions

1. **Orientation-conditioned priors enable ground truth posteriors**: Extended the model in Orbán et al. (2016) with $p(\mathbf{y}|\theta)$ to compute both $p(\theta|\mathbf{x})$ and $p(d|\mathbf{x})$, providing supervision signals for decoder evaluation.
2. **Biologically-constrained temporal data generator**: Developed hierarchical ancestral sampling (Algorithms 1-2) respecting ESS=3 limitation from calcium imaging, generating synthetic neural data with known ground truth.
3. **Systematic comparison of decoding strategies**: Demonstrated that temporal decoding outperforms spatial decoding for perceptual uncertainty (KL: 0.103 vs 0.128), with advantage specific to variance estimation (r=0.928 vs 0.736). Unexpectedly, spatial decoding (on temporally encoded synthetic data) excels for decision uncertainty - a finding requiring further investigation.
4. **Preliminary framework for true spatial encoding**: Laid the groundwork to extend to a DDC framework with 1024-dimensional nonlinear projections computing $\mathbb{E}[\sigma(\mathbf{w}^T\mathbf{y} + b)]$, establishing methodology for testing genuine spatial codes beyond temporal averaging.

2 Background and Literature Review

2.1 Bayesian Ideal Observer: Perceptual vs Decision Uncertainty

In this report, we focus on the formation and representation of perceptual and decision uncertainty in the case where it arises from visual perception (the sensory stimulus is visual). Whilst both types of uncertainty are a direct result of information limitations, they pose distinct computational challenges which might be processed differently by neural systems. Perceptual uncertainty can be defined as the uncertainty that arises as result of physical properties of the stimulus - in the context of the grating orientation task, the perceptual uncertainty can be seen as the posterior distribution over the orientation of the grating given noisy visual information (pixels on the retina), which may be further

disambiguated by various extrinsic (not a function of the visual system) nuisance variables. Decision uncertainty, on the other hand, refers to uncertainty that forms over the action taken given the visual stimulus - once again in the context of the behavioural experiment, the mouse needs to make a discriminatory effort based on the perceived orientation and the uncertainty would be over whether to categorise a grating orientation into left or right based on a vertical decision boundary. An interesting case forms when the ground truth stimulus orientation is on the left/right decision boundary - in this scenario maximum uncertainty will form over the decision as the brain has to map continuous sensory information onto the discrete behaviour categories.

Now having defined the two types of uncertainty that will be analysed in this report, we turn to using a Bayesian ideal observer framework to formalise both how they arise and how they are processed. The ideal observer provides us with a computational model for optimal statistical inference given sensory visual data, and will serve as suitable analogue of how V1 processes stimuli. In the context of our grating task, the perceptual posterior distribution over the orientation given sensory inputs is constructed as the likelihood of observing a particular sensory input given different orientations with prior knowledge about orientation statistics, which is part of the internal model of the brain:

$$p(\text{orientation}|\text{sensory input}) \propto p(\text{sensory input}|\text{orientation})p(\text{orientation})$$

The **variance of this posterior distribution** gives us a quantification of the uncertainty, with broader distributions indicating higher uncertainty about the true state of the world (true grating orientation). For decision uncertainty, which is downstream of perception in the causal inference model (inversion of generative model), the ideal observer must compute a posterior probability over each class by marginalising the perceptual posterior over orientations belonging to each category. Decision uncertainty can then be quantified by the entropy of this binary distribution, with maximum uncertainty occurring when both choices are equally probable.

We perform task related analysis and computations on this Bayesian ideal observer model in Section 3.3 and 3.4 but it is a studied concept that the V1 maintains separate representations of perceptual and decision uncertainty, giving foundation to the hierarchical processing scheme [7]. Such separation of perceptual uncertainty encoding from decision context enables sensory areas to maintain task support flexibility without needing to reprocess sensory representations.

2.2 Spatial Encoding and Decoding

With the understanding that humans compute posterior distributions over perception and decision in biological neural models, we look to the two main hypotheses that have been proposed to give us the neural representation of this uncertainty.

Probabilistic Population Codes (PPCs): In the spatial encoding scheme, the pattern of activity across a population of neurons at a single moment in time encodes an entire

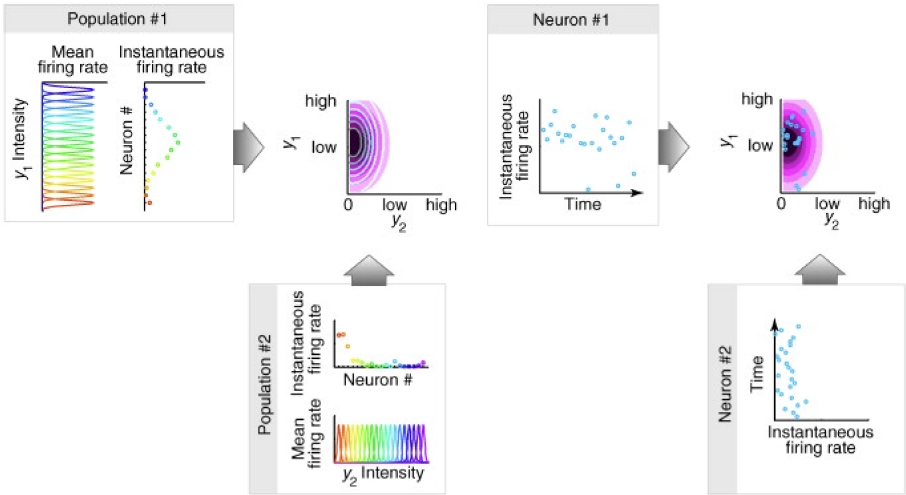


Figure 2: *Left*: Spatial Coding Scheme, *Right*: Temporal Coding Scheme. Diagram taken from Fiser 2010 [6].

probability distribution over the encoded variable. For example, let us consider a vector $\mathbf{r}$, which contains a set of activations, with each activation's value indexed by r_i. This scheme states that $\mathbf{r}$ can collectively represent more information than any individual component r_i, hence the name *spatial encoding* - information is encoded spatially across the neural population vector. The spatial coding scheme was formalised by Ma et al.[3] who established probabilistic population codes (PPCs) where the activity pattern of a neural population can encode a probability distribution. Ma et al. show that when neural variability follows Poisson-like statistics and neurons have bell-shaped tuning curves, the population response to a stimulus naturally encodes the posterior distribution through Bayes' rule:

$$p(s|\mathbf{r}) \propto p(\mathbf{r}|s)p(s) \tag{1}$$

Specifically, for independent Poisson variability with tuning curves $f_i(s)$, this becomes:

$$p(s|\mathbf{r}) \propto \prod_i \frac{e^{-f_i(s)} f_i(s)^{r_i}}{r_i!} p(s) \tag{2}$$

As shown in Figure 2 (left), each neuron represents a basis function and by considering the activity of the population, the brain could perform Bayesian inference through a linear combination (weighted summation) of these distributed representations. While PPCs show how spatial coding schemes can facilitate probabilistic computation, they are mathematically constrained to represent distributions within the exponential family - essentially limiting the framework to well-behaved distributions.

Distributed Distributional Codes (DDCs): More recent work by Vértes and Sahani[8] introduced Distributed Distributional Codes (DDCs) as a more flexible extension of probabilistic population codes whilst still allowing rapid readout of distributional

information from instantaneous population activity. Although PPCs show how neural activities can directly serve as sufficient statistics for the posterior distribution, a critical limitation is that the representable distributions are limited to the exponential family. DDCs overcome this limitation through a two-stage encoding where neural activities represent expected values of K non-linear encoding functions, $r^{(i)}(x, \phi) \approx \langle T^{(i)}(\mathbf{y})\rangle_{p_\theta(\mathbf{y}|x)}$. These expectations then serve as sufficient statistics for a maximum entropy distribution:

$$q(\mathbf{y}|x) = \frac{1}{Z(\eta(x))} \exp\left(\sum_{i=1}^{K} \eta^{(i)}(x) T^{(i)}(\mathbf{y})\right) \tag{3}$$

where the encoding functions take the form $T^{(i)}(\mathbf{y}) = \sigma(\mathbf{w}^{(i)T}\mathbf{y} + b^{(i)})$ for $i = 1...K$. For example, in the case where we have the first and second moments, the max entropy distribution will be Gaussian - but we are not limited to any specific family and DDCs provide us with a way to represent the complex, potentially non-Gaussian posterior distributions.

The choice of K, our number of encoding functions, requires consideration as it involves a trade-off between representational capacity and computational efficiency. By choosing a rich enough set of encoding functions, the DDC representation can in principle capture the posterior arbitrarily well as shown by Equation (4). Unlike the PPC decoding, we never learn the full posterior, but instead we could, for example, learn the weights α during decoding.

$$f(\mathbf{y}) \approx \sum_i \alpha^{(i)} T^{(i)}(\mathbf{y}) \Rightarrow \langle f(\mathbf{y})\rangle_{q(\mathbf{y})} \approx \sum_i \alpha^{(i)} r^{(i)} \tag{4}$$

In the limit of an infinite set of embedding functions, the DDC representation approaches a mean embedding of the posterior distribution in a reproducing kernel Hilbert space. This theoretically guarantees the approximation quality of the DDC as $K \to \infty$ and links the approach to the existing work on kernel-based distribution representations[14].

2.3 Temporal Encoding and Decoding

The temporal encoding of uncertainty departs from the static approach of the spatial scheme. In temporal coding schemes, neural activity over time encodes uncertainty. The neural sampling hypothesis formalises this idea - it proposes that neural circuits do not compute with probability distributions directly but instead generate samples from these distributions, with the temporal sequence of activity patterns reflecting the underlying uncertainty. In this scheme trial-to-trial variability in cortical neurons is not mere noise but may reflect the sampling of an internal model's uncertainty[4] - this is unlike spatial schemes, where noise is seen as a nuisance variable and a function of the visual system rather than a factor that is used in the encoding of uncertainty. The temporal coding scheme doesn't encode parameters of probability distributions as seen in the PPC framework or essential statistics as done by DDCs. Instead, complex distributions are represented through samples rather than analytical forms. Hence under this framework the brain performs Monte Carlo sampling to estimate posteriors. The sampling hypothesis is specifically applied to V1 by Orbán et al.[4] and several key predictions are made:

neural variability should be stimulus-dependent and carry information about perceptual uncertainty; the temporal statistics of neural activity should match those expected from sampling from the posterior distribution; spontaneous activity should reflect the prior distribution in the absence of sensory input; and the convergence of temporal averages to the posterior mean should follow predictable dynamics. Implementing temporal codes computationally that reflect the processes in neural circuits requires methods that can generate appropriate stochastic structure whilst still allowing for inference. In this project we extend the approaches proposed by Orbán et al.[4], where samples are generated from the actual posterior, which we obtain through a combination of closed form and numerical integrals - this is explored in later sections.

Unlike spatial codes, which can be read out instantaneously, temporal codes require different decoding approaches. We construct empirical distributions from the sequence of activity patterns, $\mathbf{y}$, treating each pattern as a sample from the underlying distribution. The following section develops a clearer understanding of the temporal framework's computations that we use in this project.

2.4 The Neural Representation of Perceptual Uncertainty in the Visual Cortex

To simulate uncertainty in V1, we require a generative model that can generate both sensory data and can then be inverted in order to compute the posteriors over the stimulus of interest. We base our generative model on the work of Orban 2016 [4], who proposed a *Gaussian Scale Mixture* (GSM) model, which assumes that an image can be represented by a set of underlying latent feature coefficients that are subject to a global random scaling (contrast) and additive noise.

Mathematically, we can write:

$$\mathbf{x} = z\, A\, \mathbf{y} + \boldsymbol{\epsilon} \tag{5}$$

where

- $\mathbf{x} \in \mathbb{R}^{D_x}$ is the observed image (a vector of pixel intensities within a patch of the visual field).
- A is a $D_x \times D_y$ matrix whose columns are basis functions (features). In our context, the basis A can be thought of as a set of Gabor filters or receptive fields similar to those in V1. The matrix A maps the latent feature space to the image space.
- $\mathbf{y} \in \mathbb{R}^{D_y}$ is a vector of latent feature activations. These represent how strongly each basis function is present in the image. In Orban *et al.*'s model, these latent features correspond to oriented and scaled local structure in the image.
- z is a scalar latent variable representing the *contrast scaling* (or more generally, a multiplicative gain on the features). It is a positive random variable and we assume z has a Gamma distribution.
- $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma_X^2 I)$ is additive Gaussian noise in the image (with variance σ_X^2 per pixel). This accounts for observation noise.

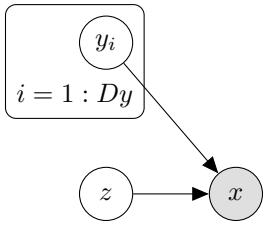


Figure 3: Graphical model of the GSM generative process. The observed image $\mathbf{x}$ is generated from latent neural features y_i ($i = 1$ to D_y) scaled by contrast variable z according to Equation (5)

Equation (5) in combination with Figure 3 captures the way that we model the mechanistic process of image generation in V1: namely, image $\mathbf{x}$ is generated by first sampling a global contrast z from a gamma prior and then using this to scale sampled latent feature coefficients $\mathbf{y}$. We then use the filter bank A to project to image space and finally sample additive noise. The concept of using Gabor filters as a computational analogue to the population of V1 neurons is a well-researched idea [13] - Gabor functions are sinusoidal waveforms modulated by a Gaussian envelope, making them adept at capturing both frequency and spatial information. They closely resemble the response profiles of V1 neurons to edges, orientation and scale whilst providing a compact and efficient way to model complex visual features.

In the original Orban et al. (2016) framework, the generative model employed a simple unconditional prior for the latent features:

$$p(\mathbf{y}) \sim \mathcal{N}(\mathbf{0}, \mathbf{C}) \tag{6}$$

This zero-mean Gaussian represents the brain's internal model of natural image statistics, with covariance structure learned from image data. However, for our orientation discrimination task, we will require a more structured latent space that explicitly captures how orientation information is encoded in these features (see Section 3.2).

By inverting the generative process, Orbán introduces the inference process, with the objective being to determine the posterior distribution $p(\mathbf{y}|\mathbf{x})$. We can first obtain a closed form posterior over z,

$$p(z \mid \mathbf{x}) \propto p(z)\, \mathcal{N}\Big(\mathbf{x}; \mathbf{0}, z^2\mathbf{A}\mathbf{C}\mathbf{A}^\top + \sigma_x^2\mathbf{I}\Big) \tag{7}$$

and then condition on z to obtain a posterior distribution over feature activations $\mathbf{y}$:

$$p(\mathbf{y} \mid z, \mathbf{x}) = \mathcal{N}\Big(\mathbf{y}; \boldsymbol{\mu}(z, \mathbf{x}), \boldsymbol{\Sigma}(z)\Big) \tag{8}$$

The posterior mean and covariance of feature activations are given by:

$$\boldsymbol{\Sigma}(z) = \left(\mathbf{C}^{-1} + \frac{z^2}{\sigma_x^2}\mathbf{A}^\top\mathbf{A}\right)^{-1}, \quad \boldsymbol{\mu}(z, \mathbf{x}) = \frac{z}{\sigma_x^2}\boldsymbol{\Sigma}(z)\mathbf{A}^\top\mathbf{x}.$$

We can then marginalise over the contrast variable z to obtain the posterior as shown. This resulting model provides us with the crucial foundation to generate synthetic neural data for the temporal encoding.

$$p(\mathbf{y}|\mathbf{x}) = \int p(z|\mathbf{x})p(\mathbf{y}|z,\mathbf{x})\mathrm{d}z. \tag{9}$$

2.5 A Data-Driven Approach for Testing Specific Hypotheses About Probabilistic Representations

The experimental approach to explore the neural coding hypothesis was presented by Amvrosiadis [9] and Koblinger [10], and we closely follow the experimental paradigm in our computational modelling methodology (Section 3). In their work, the authors use feed forward neural networks (termed as *neural decoders*) as flexible decoders to test the different coding hypotheses about uncertainty representation. Here an important distinction must be made - these networks serve purely as analytical tools - they are not intended as models of biological computation, but rather as powerful function approximators capable of learning complex mappings from neural activity patterns to probability distributions. This approach is agnostic about the exact form of the neural code. Instead of assuming specific encoding functions, parametric forms or mechanisms, we let the networks discover whatever transformations best map from neural activity to uncertainty. Following the computational methods of Koblinger et al. by training separate decoders under different assumptions about the neural code (temporal vs. spatial) and comparing their performance, we can test different coding hypotheses.

For the neural decoders, the Kullback-Leibler (KL) divergence between decoded and behaviourally-derived posteriors is the training objective. The research showed that temporal coding schemes consistently outperform spatial coding schemes in capturing uncertainty. Specifically, decoders that integrate information across multiple time bins (treating neural activity as samples) better match the posteriors than decoders operating on instantaneous (averaged) activity patterns [10]. Further analysis such as using individual time bins performed worse than integration, supporting the sampling hypothesis (temporal code) and the idea that variability isn't just noise - if this was the case, averaging samples to single time bins would improve signal to noise ratio.

Imaging techniques were another significant consideration in the behavioural experiment - calcium imaging is widely used for studying population coding in V1 as it allows researchers to simultaneously record from hundreds of neurons with cellular resolution. Koblinger et al. use GCaMP6s, which provides a fluorescent readout of intracellular calcium concentration, reflecting neural spiking activity. However, such calcium imaging techniques have some temporal resolution constraints, which will affect the computational methodologies used by the authors and therefore adopted in this project. The Effective Sample Size (ESS) quantifies how temporal resolution limits sampling-based codes. In the context of the orientation discrimination experiment, autocorrelation analysis of neural responses during the 1-second stimulus presentation period reveals an ESS of approximately 3 [10]. Although the imaging was done continuously during stimulus presentation, the temporal correlation structure of the calcium imaging limited the number of samples that could be extracted

whilst ensuring samples were independent and identically distributed (i.i.d.).

3 Computational Framework and Model Development

The gap between theoretical proposals for neural codes of uncertainty and empirical tests of these theories remains challenging. We now develop methods of synthetic data generation coupled with neural decoders trained on specific hypotheses. This will build towards providing researchers with computational tools to examine the competing theories on real biological data.

The synthetic data generation in this project begins with the generative model introduced by Orbán et al.[4], and analysed in Section 2.4. Here, we sample $\mathbf{y}$ from the prior defined by Equation (6), and then a complete generative process would involve sampling the scale variable, z, from a Gamma prior and constructing the image as per Equation (5). But in this initial study we choose to control contrast and later aperture of images, $\mathbf{x}$, to build an understanding of the applicability of this generative and inference model as a foundation for our use cases.

3.1 Model Validation: Effects of Contrast and Aperture

Sensory uncertainty arises as different world states are capable of producing similar and in many cases indistinguishable sensory inputs. As such the brain only ever has access to noisy measurements as opposed to the true world state, and it must be capable of operating within this regime of uncertainty. We introduce in Section 1.1 the concept of the Bayesian brain, which is a well-established framework under which animal brains may process and optimally handle this uncertainty [11]. The brain maintains a prior distribution reflecting the internal model of the world and upon receiving sensory information it computes a posterior in a Bayesian way, encapsulating the spread of possible states weighted by probability [2] allowing for a full representation of uncertainty. Bayesian behaviour is well documented in psychophysics and neurobiology - humans combine cues in a way that is close to Bayesian optimal, weighting sensory evidence by its reliability [12].

The Primary visual cortex (V1), the first cortical area to process visual input, forms the testing ground for our study on uncertainty and to modulate uncertainty in V1, we control two nuisance variables:

- **Contrast**: the difference in luminance or colour that makes an object distinguishable from the background. Low- contrast images (e.g., faint or washed-out stimuli) provide weaker evidence and thus induce greater uncertainty about features like orientation.
- **Aperture**: the portion of the visual field or the size of the viewing window. A small aperture (or a occluded view) means only part of the object is seen, again increasing uncertainty since multiple interpretations of the limited view are possible.

To understand the effects of the experiment's nuisance variables on our computational

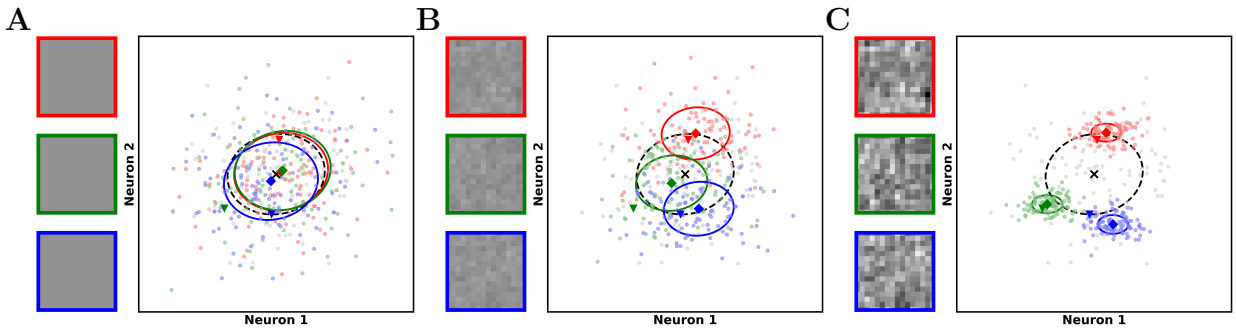


Figure 4: Posterior inference of neural responses under varying stimulus contrast levels. (A) Zero contrast, **(B)** intermediate contrast, **(C)** high contrast. Left insets show three 16×16 pixel stimulus images with coloured borders (red, green, blue) corresponding to three independent samples from the generative model. Right panels display the posterior distributions over two example neurons in response to each stimulus. Black dashed ellipse indicates the prior distribution $p(\mathbf{y})$ with gray dots showing prior samples. Coloured ellipses represent posterior covariances $p(\mathbf{y}|\mathbf{x})$ for each stimulus, with corresponding posterior samples (light coloured dots). As contrast increases, posterior distributions become more concentrated around the true values of $\mathbf{y}$.

model, we look at reproducing some findings of Orbán et al. (2016) [4]. With the original code base from this foundational paper being in Matlab, the first task was to convert the inference functions into Python. To verify the Python implementation, key results from the paper were reproduced, focusing on the effects of contrast on the uncertainty of latent variables. By generating artificial stimuli and modulating contrast systematically, this formed a reliable sanity check of the code base. We performed inference and Figure 4 shows plots of the sampling activity of two neurons for three generated images at increasing levels of contrast.

From a Bayesian perspective the uncertainty over an inferred quantity (orientation of grating) depends on the nuisance variables: if contrast is low or aperture is limited the posterior will broaden representing increased uncertainty. Perceptual inference marginalises out (integrate) these nuisance variables to compute the quantity of interest, which in our case is the grating orientation. A central premise of the Bayesian brain hypothesis is neurons in V1 might not only signal features like orientation and spatial frequency of the grating, but also how confident or uncertain the network is about these features [11]. As the matrix A used in the Orbán generative process encodes Gabor filters (scaled and oriented wavelets) it is inherently orientation selective and to study how this selectivity manifests in the posterior distributions, we generate oriented gratings as per Appendix A.2 and vary contrast and aperture to form the tuning curves seen in Figure 5.

3.2 Extending the Model: Orientation-Conditioned Priors

Having verified orientation selectivity of our current inference process, we extend this to the context of our specific task. The **stimulus orientation** θ is a (discrete) latent variable that influences $\mathbf{y}$. We assume that $\mathbf{y}$ is drawn from an orientation-conditioned

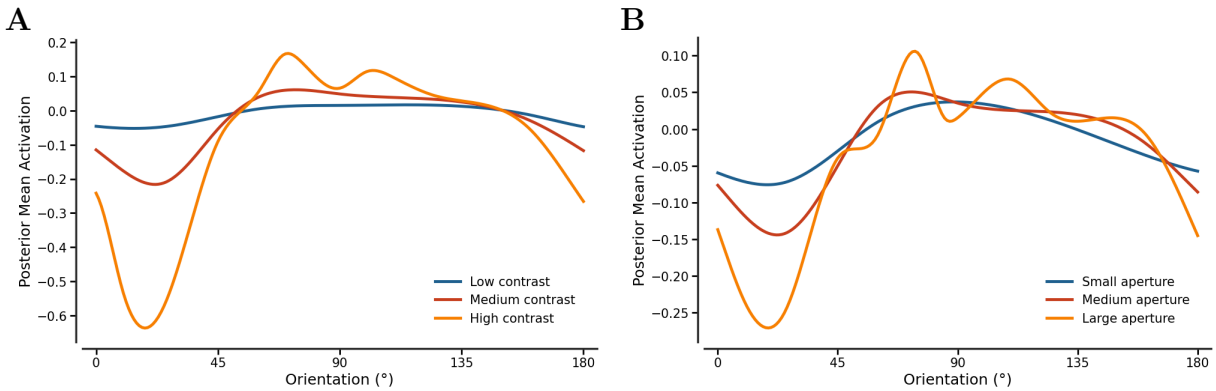


Figure 5: Exploring orientation selectivity of a particular latent neural feature y_i as we systematically vary nuisance factors. (**A**) Tuning curves showing posterior mean activation as a function of orientation (0 - 180°) for three different contrast levels. Low contrast (blue) produces relatively flat tuning with minimal orientation selectivity, while medium (red) and high contrast (orange) stimuli elicit stronger orientation-selective responses with pronounced peaks and troughs. (**B**) Tuning curves for three different aperture sizes with fixed contrast. Small apertures (blue) yield modest orientation tuning, while medium (red) and large apertures (orange) produce more peaked tuning curves. Together, these results demonstrate how the same basis functions can exhibit varying degrees of orientation selectivity depending on stimulus parameters.

prior distribution and extend the Orbán model by including θ as a latent variable that parametrises the distribution of $\mathbf{y}$. Let θ denote the true orientation of the grating stimulus (we assume $\theta \in 1°, 2°, \ldots, 180°$ for convenience of representation as 180 classes). We treat θ as a random variable with a prior $p(\theta)$ (which for our simulations we take to be uniform over the chosen set of orientations). Now we modify the prior over $\mathbf{y}$ as follows:

$$p(\mathbf{y} \mid \theta) \approx \mathcal{N}\Big(\mathbf{y};\, \boldsymbol{\mu}_\theta,\, C_\theta\Big). \tag{10}$$

That is, conditioned on a particular orientation θ, the latent features $\mathbf{y}$ are Gaussian with mean vector $\boldsymbol{\mu}_\theta$ and covariance matrix C_θ. This formulation allows the model to capture the notion that if the stimulus orientation is a grating at 45°, then the latent feature vector $\mathbf{y}$ is likely to take certain patterns (for example, the coefficients corresponding to basis functions aligned with 45° might have higher mean activation). The original model [4] did not have an explicit θ variable. Under our extended model, with uninformative stimuli (for example a low contrast setting where the stimulus provides no sensory data), our orientation conditioned prior would collapse back to the original Orbán model's prior where we set $\boldsymbol{\mu}_\theta = 0$ for all θ and just have $C_\theta = C$ independent of θ. But given orientation information is central to our task, the original prior does not allow orientation to systematically affect the latent distribution beyond what A can impose. Thus the modification to the prior introduced by Equation (10) was mathematically necessary as it allows us to compute the posterior $p(\theta|\mathbf{x})$ by Bayes' rule - we need to be able to model how latent features are distributed given a specific orientation hypothesis as this forms the basis of our study on perceptual uncertainty.

To actually form the conditional priors as an extension from the existing parameters,

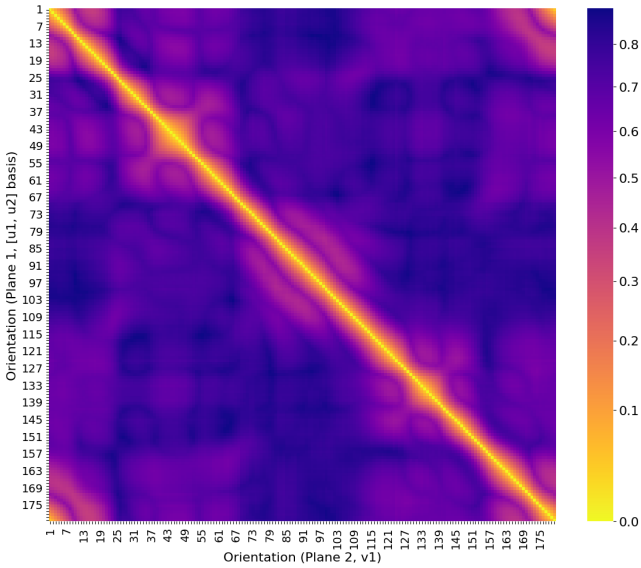


Figure 6: Pairwise Reconstruction Error shows how well each orientation's two-dimensional principal-component subspace can reconstruct the principal components of every other orientation. Lower values (warmer colours) indicate stronger similarity between the latent representations of those two orientations, while higher values (cooler colours) indicate that their latent subspaces differ.

the fitting process involved generating thousands of synthetic grating images for each orientation (1–180°) with a set of varying contrast and aperture conditions, and then using the inference engine (Section 2.4) to sample latent vectors $\mathbf{y}$ from the posterior $p(\mathbf{y}|\mathbf{x})$ for each image. By aggregating these samples across images of the same orientation, we efficiently estimated the orientation-conditioned moments:

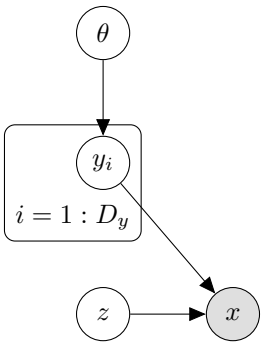
$$\mu_\theta = \frac{1}{N}\sum_{k=1}^{N}\mathbf{y}_k, \qquad C_\theta = \frac{1}{N}\sum_{k=1}^{N}\Big(\mathbf{y}_k - \mu_\theta\Big)\Big(\mathbf{y}_k - \mu_\theta\Big)^T$$

Now the full generative model is specified by

$$p(\theta)\,, \quad p(\mathbf{y} \mid \theta) = \mathcal{N}\Big(\mathbf{y};\, \boldsymbol{\mu}_\theta,\, C_\theta\Big)\,, \quad p(z) = \mathrm{Gamma}(z;\, k_\Gamma, \theta_\Gamma)\,, \quad p(\boldsymbol{\epsilon}) = \mathcal{N}\Big(\boldsymbol{\epsilon};\, 0,\, \sigma_X^2 I\Big),$$

together with the observation Equation (5).

A critical validation question was whether these fitted priors actually captured orientation-specific structure in the latent space. To establish uniqueness driven by orientation in $\mathbf{y}$ space, a reconstruction error analysis was done. We computed the principal components of each orientation's distribution and measured how well the subspace defined by one orientation could represent another orientation's primary mode of variation. The diagonal structure in the error matrix shown in Figure 6 confirms that nearby orientations share similar latent representations, whilst orientations separated by large angles incur larger reconstruction error. A full derivation of the mathematics used to conduct this analysis is available in the Appendix A.1.



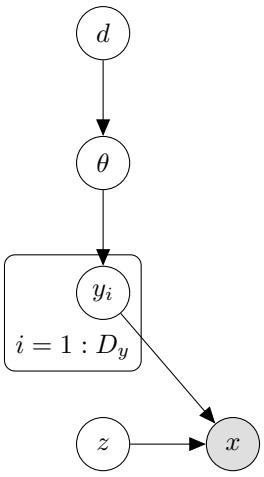


Figure 7: **Left:** extended network model showing the complete image generation process. The grating orientation θ determines the distribution of latent features y_i, which together with contrast variable z generate the observed image $\mathbf{x}$.

Figure 8: **Right:** extended network model showing the complete image generation process with decision variable. The decision variable d determines the distribution of orientation θ, which influences the latent features y_i. Together with contrast variable z, these generate the observed image $\mathbf{x}$.

3.3 Bayesian Ideal Observer Model For Perceptual Uncertainty

Having established our orientation-conditioned priors, we formalise the perceptual inference framework by constructing a Bayesian ideal observer model as an inversion of the $\theta-$extended generative model seen in Figure 7. This generative model illustrates the hierarchical structure of visual processing and its inversion serves as our computational analogue for how an optimal observer would process visual information, providing ground truth posteriors over orientation, against which we can evaluate different neural coding schemes.

To begin our derivation, we first express the complete joint distribution over all variables in our model. This joint distribution shows the graphical dependency structure between orientation θ, nuisance variables z (which encapsulate contrast and aperture), neural latent features $\mathbf{y}$ and the observed image $\mathbf{x}$.

$$p(\mathbf{x},\, \mathbf{y},\, z,\, \theta) \;=\; p(\mathbf{x} \mid \mathbf{y},\, z,\, \theta)\, p(\mathbf{y} \mid \theta)\, p(z)\, p(\theta). \tag{11}$$

This factorisation importantly reflects some key modelling assumptions that will be used in further derivations: namely, the image depends on the latent activations $\mathbf{y}$ and nuisance variables z, the latent activations are determined by the orientation, and both contrast and orientation have independent priors.

Expressing $p(\mathbf{y} \mid \mathbf{x})$ via marginalisation: The key distribution of interest is the posterior over the latent activations vector $\mathbf{y}$, as it is this distribution that will later enable us to construct a temporal neural data generator. Given an observed image (grating stimuli) $\mathbf{x}$, and in the case where z is continuous and θ is discrete, we can write this in a factorised form as shown in Equation (12). Hence, if we can compute each of $p(\theta \mid \mathbf{x})$, $p(z \mid \mathbf{x}, \theta)$, $p(\mathbf{y} \mid \mathbf{x}, z, \theta)$, then sampling from $p(\mathbf{y} \mid \mathbf{x})$ is straightforward and can be utilised in the temporal scheme.

$$p(\mathbf{y} \mid \mathbf{x}) \;=\; \sum_{\theta} \int p\Big(\mathbf{y},\, z,\, \theta \mid \mathbf{x}\Big)\, \mathrm{d}z = \sum_{\theta} \int p\Big(\mathbf{y} \mid z,\, \theta,\, \mathbf{x}\Big)\, p\Big(z \mid \theta,\, \mathbf{x}\Big)\, p\Big(\theta \mid \mathbf{x}\Big)\, \mathrm{d}z \tag{12}$$

Deriving the perceptual posterior, $p(\theta \mid \mathbf{x})$: This is the posterior that represents our ground truth perceptual uncertainty (the posterior distribution over the grating orientation), and it will form the supervision signal for our neural decoders in subsequent sections, making its accurate computation critical to our methodology.

Computing this posterior requires us to evaluate the likelihood of observing image $\mathbf{x}$ under each possible orientation hypothesis. However, the relationship between orientation and image is mediated by both the neural features and contrast variable, necessitating a two-stage marginalisation process. For each candidate orientation θ, we must first marginalise over all possible contrast values:

$$p(\mathbf{x} \mid \theta) \;=\; \int p(\mathbf{x} \mid z, \theta)\, p(z)\, \mathrm{d}z \tag{13}$$

The inner likelihood itself requires marginalising over the neural features i.e. $p(\mathbf{x} \mid z, \theta) = \int p(\mathbf{x} \mid \mathbf{y}, z, \theta)\, p(\mathbf{y} \mid \theta)\, \mathrm{d}\mathbf{y}$. Fortunately, given our Gaussian assumptions, this marginalisation can be performed analytically. The key insight is that when we integrate out the Gaussian-distributed neural features, the resulting distribution over images remains Gaussian:

$$p(\mathbf{x} \mid z, \theta) \;=\; \mathcal{N}\Big(\mathbf{x};\; z\, A\, \boldsymbol{\mu}_{\boldsymbol{\theta}},\; \sigma_X^2 I + z^2\, A\, C_\theta\, A^T\Big) \tag{14}$$

This result has an intuitive interpretation: the mean of the image distribution is determined by projecting the orientation-specific mean neural activity through the basis functions, scaled by contrast. The covariance combines observation noise with the projected uncertainty in neural features, also scaled by contrast.

In practice, the continuous integral over contrast must be approximated numerically. We discretise the contrast variable over a grid $\{z_j\}_{j=1}^{D_z}$ with spacing Δz, chosen to capture the relevant range of the Gamma prior while maintaining computational efficiency:

$$p(\mathbf{x} \mid \theta) \approx \Delta z \sum_{j=1}^{D_z} p(\mathbf{x} \mid z_j, \theta)\, p(z_j). \tag{15}$$

With these orientation-specific likelihoods computed, we can apply Bayes' rule to obtain

the posterior. Assuming a uniform prior over orientations (reflecting no initial bias toward any particular orientation), the posterior simplifies to a normalised version of the likelihood:

$$p(\theta \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid \theta)\, p(\theta)}{\sum_{\theta'} p(\mathbf{x} \mid \theta')\, p(\theta')} \propto p(\mathbf{x} \mid \theta) \tag{16}$$

Substituting our numerical approximation yields the final computational formula for the orientation posterior:

$$p(\theta \mid \mathbf{x}) \approx \frac{\Delta z \sum_{j=1}^{D_z} p(\mathbf{x} \mid z_j, \theta)\, p(z_j)}{\Delta z \sum_{\theta'} \sum_{j=1}^{D_z} p(\mathbf{x} \mid z_j, \theta')\, p(z_j)} = \frac{\sum_{j=1}^{D_z} p(\mathbf{x} \mid z_j, \theta)\, p(z_j)}{\sum_{\theta'} \sum_{j=1}^{D_z} p(\mathbf{x} \mid z_j, \theta')\, p(z_j)} \tag{17}$$

This posterior captures how perceptual uncertainty varies with stimulus conditions. When contrast is low or aperture is limited, the likelihood functions become broader and more similar across orientations, leading to a flatter posterior that reflects increased uncertainty.

Deriving $p(z \mid \mathbf{x}, \theta)$: Having computed the orientation posterior, we now turn to the contrast posterior conditioned on a specific orientation hypothesis. Starting from Bayes' rule, the posterior over z for a fixed orientation θ is given by

$$p(z \mid \mathbf{x}, \theta) \propto p(\mathbf{x} \mid z, \theta)\, p(z) \tag{18}$$

The likelihood $p(\mathbf{x} \mid z, \theta)$ is obtained analytically as before. The prior over z is a Gamma distribution given by $p(z) = \text{Gamma}(z; k_\Gamma, \theta_\Gamma)$. To compute the posterior, we discretise z over a grid as before, and for each point we evaluate the unnormalised posterior $\tilde{p}(z_j \mid \mathbf{x}, \theta) = p(\mathbf{x} \mid z_j, \theta)\, p(z_j)$ which we then normalise to obtain the continuous posterior density approximated at the grid points:

$$p(z \mid \mathbf{x}, \theta)\Big|_{z=z_j} \approx \frac{p(\mathbf{x} \mid z_j, \theta) p(z_j)}{\Delta z \sum_{k=1}^{D_z} p(\mathbf{x} \mid z_k, \theta) p(z_k)} \tag{19}$$

Deriving $p(\mathbf{y} \mid \mathbf{x}, z, \theta)$: The final component of our inference framework is the posterior over neural features conditioned on both orientation and contrast. This distribution represents the immediate neural response to a stimulus under a specific interpretation of its orientation and contrast properties.

Given our linear-Gaussian generative model, this posterior takes a particularly elegant form. The conditioning on specific values of z and θ reduces our model to a standard linear-Gaussian system:

$$\mathbf{x} \;=\; z\, A\, \mathbf{y} + \boldsymbol{\epsilon}, \quad \mathbf{y}|\theta \sim \mathcal{N}(\boldsymbol{\mu_\theta},\, C_\theta), \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0,\, \sigma_X^2 I).$$

Standard results from multivariate Gaussian theory immediately give us the posterior distribution shown in Equation (20) and we have therefore obtained all the distributions in the factorisation of $p(\mathbf{y}|\mathbf{x})$ (Equation (12)). This forms the foundation for our temporal coding implementation, where neural responses are modelled as samples from this posterior distribution.

$$p(\mathbf{y} \mid \mathbf{x}, z, \theta) \;=\; \mathcal{N}\Big(\mathbf{y};\; \boldsymbol{\mu_z},\; \Sigma_z\Big) \tag{20}$$

where

$$\Sigma_z^{-1} \;=\; C_\theta^{-1} \;+\; \frac{z^2}{\sigma_X^2}\, A^T\, A \tag{21}$$

$$\boldsymbol{\mu_z} \;=\; \Sigma_z\Big(C_\theta^{-1}\, \boldsymbol{\mu_\theta} \;+\; \frac{z}{\sigma_X^2}\, A^T\, \mathbf{x}\Big) \tag{22}$$

3.4 Extending the Bayesian Ideal Observer Model to the Decision Variable

The framework developed thus far addresses perceptual uncertainty - how the visual system represents uncertainty about continuous stimulus properties like orientation. However, the behavioural task in our experimental paradigm requires a further computational step. The experimental set-up described in Section 1.2 involved a perceptual decision-making task, in which the actual outcome observed by the researcher is when the mice map the stimuli to discrete responses. Specifically, in the orientation discrimination paradigm mice need to categorise the oriented gratings into “left” (0–90°) versus “right” (90–180°). To model this process, we extend our Bayesian ideal observer framework to include a binary decision variable $d \in \{-1, +1\}$, representing these categorical choices. The extended generative model now incorporates an additional level in the hierarchy as shown in Figure 8.

The joint distribution over all variables in this extended model reflects this additional hierarchical structure:

$$p(d, \theta, z, \mathbf{y}, \mathbf{x}) = p(\mathbf{x}|\mathbf{y}, z)p(\mathbf{y}|\theta)p(z)p(\theta|d)p(d) \tag{23}$$

The prior over the decision variable d follows a Bernoulli distribution:

$$p(d) = \begin{cases} \rho & \text{if } d = -1 \\ 1 - \rho & \text{if } d = +1 \end{cases} \tag{24}$$

where we initially set $\rho = 0.5$, reflecting an unbiased observer. This is a valid assumption under theoretical settings for an ideal observer and a complete training set with equal number of images at all orientations (or decisions). However, we could also choose to fit ρ: one method would be estimating ρ from experimental data using maximum likelihood estimation and setting it to the empirical frequency of *left* decisions. Alternatively, a

Bayesian approach could be adopted if we have informative prior beliefs about response biases, allowing us to integrate previous knowledge about a participant's decision tendencies with observed choices.

The conditional distribution of orientations given a decision enforces the categorical structure of the task:

$$p(\theta|d) = \begin{cases} \frac{1}{n_{-1}} & \text{if } d = -1 \text{ and } \theta \in [0°, 90°) \\ \frac{1}{n_{+1}} & \text{if } d = +1 \text{ and } \theta \in [90°, 180°) \\ 0 & \text{otherwise} \end{cases} \tag{25}$$

where n_{-1} and n_{+1} are the number of discrete orientations in each category. This distribution assigns uniform probability to orientations consistent with the decision and zero probability to inconsistent orientations.

The remainder of the model ($p(\mathbf{y}|\theta)$, $p(z)$, and $p(\mathbf{x}|\mathbf{y}, z)$) remains identical to the joint distribution derived for the perceptual uncertainty Bayesian Ideal Observer case from Section 3.3.

Expressing $p(\mathbf{y} \mid \mathbf{x})$ via marginalisation: Once again we start by seeking the posterior over latent features $\mathbf{y}$, which is obtained through marginalisation:

$$p(\mathbf{y}|\mathbf{x}) = \sum_d p(d|\mathbf{x}) \sum_{\theta \in \Theta_d} p(\theta|\mathbf{x}, d) \int p(\mathbf{y}|\mathbf{x}, \theta, z)\, p(z|\mathbf{x}, \theta)\, dz \tag{26}$$

With the decision variable integrated into our model, the posterior over neural features now requires an additional marginalisation step. The complete expression in Equation (26) reveals how decision uncertainty propagates through the inference hierarchy.

This marginalisation captures the full complexity of perceptual decision-making: we must consider both possible decisions, all orientations consistent with each decision, and all possible contrast values. The weighting by $p(d|\mathbf{x})$ ensures that more probable decisions contribute more strongly to the neural representation.

We need to compute several distributions that were not seen in the previous case: the posterior $p(d \mid \mathbf{x})$ and the conditional posteriors $p(\theta \mid \mathbf{x}, d)$ and $p(\mathbf{y} \mid \mathbf{x}, d)$. We also recompute the unconditional posterior over orientations, $p(\theta|\mathbf{x})$, which now requires marginalising out the decision variable d.

Deriving the decision posterior, $p(d \mid \mathbf{x})$: The posterior probability of each decision given the observed image is:

$$p(d|\mathbf{x}) = \frac{p(\mathbf{x}|d)\,p(d)}{p(\mathbf{x})} = \frac{p(d)\int p(\mathbf{x}|\theta)\,p(\theta|d)\,d\theta}{\sum_{d'} p(d')\int p(\mathbf{x}|\theta)\,p(\theta|d')\,d\theta} \tag{27}$$

For our discrete set of orientations, this integral becomes a sum:

$$p(d|\mathbf{x}) = \frac{p(d)\sum_{\theta\in\Theta_d} p(\mathbf{x}|\theta)\,p(\theta|d)}{\sum_{d'} p(d')\sum_{\theta\in\Theta_{d'}} p(\mathbf{x}|\theta)\,p(\theta|d')} \tag{28}$$

where Θ_d is the set of orientations consistent with decision d, and $p(\mathbf{x}|\theta)$ is computed as in the original model by marginalising over z:

$$p(\mathbf{x}|\theta) = \int p(\mathbf{x}|z,\theta)\,p(z)\,dz \tag{29}$$

With our uniform conditional prior $p(\theta|d) = 1/n_d$ for all $\theta \in \Theta_d$, this simplifies to:

$$p(d|\mathbf{x}) = \frac{p(d)\frac{1}{n_d}\sum_{\theta\in\Theta_d} p(\mathbf{x}|\theta)}{\sum_{d'} p(d')\frac{1}{n_{d'}}\sum_{\theta\in\Theta_{d'}} p(\mathbf{x}|\theta)} \tag{30}$$

Conditional Posterior Over Orientations For each decision d, we compute the posterior over orientations, conditioned on both the observed image and the decision:

$$p(\theta|\mathbf{x},d) = \frac{p(\mathbf{x}|\theta)\,p(\theta|d)}{p(\mathbf{x}|d)} = \frac{p(\mathbf{x}|\theta)\,p(\theta|d)}{\sum_{\theta'\in\Theta_d} p(\mathbf{x}|\theta')\,p(\theta'|d)} \tag{31}$$

With our uniform $p(\theta|d)$ within each decision category, this reduces to a renormalised likelihood over the restricted set of orientations:

$$p(\theta|\mathbf{x},d) = \frac{p(\mathbf{x}|\theta)}{\sum_{\theta'\in\Theta_d} p(\mathbf{x}|\theta')} \tag{32}$$

This conditional posterior is non-zero only for orientations consistent with the decision ($\theta \in \Theta_d$), reflecting the categorical constraint imposed by the decision process.

Unconditional posterior over orientations: We can compute the full posterior $p(\theta|\mathbf{x})$ by marginalising over decisions as per Equation (33). This marginal posterior differs subtly but importantly from the purely perceptual posterior derived in Section 3.3. While both represent uncertainty about orientation, the decision-augmented version reflects the additional structure imposed by the categorical task.

$$p(\theta|\mathbf{x}) = \sum_d p(\theta|\mathbf{x},d)\,p(d|\mathbf{x}) \tag{33}$$

Now we have all the distributions required to both form the posterior over the latents (Equation (26)), but also a perceptual posterior derived from the inversion of the complete

generative model of Figure 8.

4 Temporal Coding Implementation and Decoding Strategies

As per the definition of the temporal coding scheme (Section 2.3), we take the view that each neural response is one sample from the posterior distribution over the vector latent variables $\mathbf{y}$. Over time, multiple independent samples are drawn from this posterior, and downstream processes can consider the variation among them in order to estimate the uncertainty.

4.1 Temporal Encoding Neural Data Generator

To generate synthetic neural data under the temporal coding hypothesis, we are required to draw samples from the posterior distribution $p(\mathbf{y}|\mathbf{x})$. Rather than directly sampling from this complex distribution, we employ hierarchical ancestral sampling, which reflects the graphical model structure (Figure 7). Algorithm 1 formalises the procedure of generating samples through exploiting this graphical model. We may draw N sequential samples and under the temporal neural data generator these represent the time-varying activity patterns that encode uncertainty in V1.

Algorithm 1 Hierarchical Ancestral Sampling from $p(\mathbf{y}|\mathbf{x})$

Require: Observed image $\mathbf{x}$, number of samples N
Ensure: Samples $\{\mathbf{y}^{(1)}, \mathbf{y}^{(2)}, \ldots, \mathbf{y}^{(N)}\}$ from $p(\mathbf{y}|\mathbf{x})$
Compute $p(\theta|\mathbf{x})$ for all orientations θ using Eq. 17
for $i = 1$ *to* N **do**
  Sample $\theta^{(i)} \sim p(\theta|\mathbf{x})$
  Sample $z^{(i)} \sim p(z|\mathbf{x}, \theta^{(i)})$
  Compute $\mathbf{\Sigma}_{z^{(i)}}$ and $\boldsymbol{\mu}_{z^{(i)}}$ using eq. (21) and eq. (22)
  Ensure $\mathbf{\Sigma}_{z^{(i)}}$ is positive-definite by applying small regularisation if needed
  Sample $\mathbf{y}^{(i)} \sim \mathcal{N}(\boldsymbol{\mu}_{z^{(i)}}, \mathbf{\Sigma}_{z^{(i)}})$
end
return $\{\mathbf{y}^{(1)}, \mathbf{y}^{(2)}, \ldots, \mathbf{y}^{(N)}\}$

As an extension to the above, to generate samples from the posterior distribution once we had introduced the decision variable, we implement a hierarchical ancestral sampling procedure as before that follows the structure of the graphical model (Figure 8). A pseudo-code implementation of this procedure is shown in Algorithm 2.

Algorithm 2 Hierarchical Sampling from the Decision-Augmented Model

Require: Observed image $\mathbf{x}$, number of samples N
Ensure: Samples $\{\mathbf{y}^{(1)}, \mathbf{y}^{(2)}, \ldots, \mathbf{y}^{(N)}\}$ from $p(\mathbf{y}|\mathbf{x})$
Compute $p(d|\mathbf{x})$ for each decision $d \in \{-1, +1\}$
for $i = 1$ *to* N **do**
  Sample $d^{(i)} \sim p(d|\mathbf{x})$
  Compute $p(\theta|\mathbf{x}, d^{(i)})$ for all $\theta \in \Theta_{d^{(i)}}$
  Sample $\theta^{(i)} \sim p(\theta|\mathbf{x}, d^{(i)})$
  Compute $p(z|\mathbf{x}, \theta^{(i)})$ by discretising z and normalising
  Sample $z^{(i)} \sim p(z|\mathbf{x}, \theta^{(i)})$
  Compute $\mathbf{\Sigma}_{z^{(i)}}$ and $\boldsymbol{\mu}_{z^{(i)}}$ using eq. (21) and eq. (22)
  Ensure $\mathbf{\Sigma}_{z^{(i)}}$ is positive-definite by applying small regularisation if needed
  Sample $\mathbf{y}^{(i)} \sim \mathcal{N}(\boldsymbol{\mu}_{z^{(i)}}, \mathbf{\Sigma}_{z^{(i)}})$
end
return $\{\mathbf{y}^{(1)}, \mathbf{y}^{(2)}, \ldots, \mathbf{y}^{(N)}\}$

4.2 Temporal Decoding From a Temporal Encoding

Having developed a robust method for synthetic data generation, we now turn our focus to training neural decoders, which in the first instance will be tasked to output the perceptual uncertainty (the orientation posterior) using the sampled latent features $\mathbf{y}$ as inputs. In both the temporal and spatial decoding (Section 4.3), we implement a standard feedforward neural network with two hidden layers (512 units each with ReLU activation) followed by a 180-unit output layer with log-softmax activation, which proved sufficient to learn the mapping from 248 dimensional latent activation vectors to orientation distributions.

In the decoder training process we will use a fixed number of training samples - let $N_{\text{samples}} = 3$ be the number of samples from the posterior (generated as per the process of Algorithm 1) that we allow. This decision is driven by the actual experimental limitations following autocorrelation analysis of the data (see Section 2.5). So for each stimulus $\mathbf{x}$, the temporal coding generator will produce N_{samples} vectors: $\mathbf{y}^{(1)}, \mathbf{y}^{(2)}, \mathbf{y}^{(3)}$, each drawn i.i.d. from $p(\mathbf{y}|\mathbf{x})$, and in the first case of the temporal decoder we assume a sequential readout where the decoder will process each sample and accumulate information over time.

In our implementation of the temporal decoder, we feed each sample $\mathbf{y}^{(i)}$ through the same network and then aggregate the output probability distribution. In training we present multiple samples for each $\mathbf{x}$ and define the loss on the aggregated output. The temporal decoding rule can be summarised by Equation (34)

$$\hat{p}(\theta \mid \mathbf{x}) \;=\; \frac{1}{N_{\text{samples}}} \sum_{i=1}^{N_{\text{samples}}} f_{\text{NN}}\left(\mathbf{y}^{(i)}\right)_{\theta} \tag{34}$$

where $f_{\text{NN}}(\mathbf{y})_\theta$ denotes the probability assigned to orientation θ by the neural-network decoder given input $\mathbf{y}$. In the methodologies outlined within Section 4, we focus on the perceptual posterior, but the reader should note that the decision posterior predictions,

$\hat{p}(d|\mathbf{x})$, follow identical decoding strategies, simply replacing the 180-dimensional orientation output with a 2-dimensional decision output and changing the supervision signal to the ground truth $p(d|\mathbf{x})$. The remaining neural decoder architecture stays identical.

4.3 Spatial Decoding From a Temporal Encoding

In the spatial coding scheme, the neural population at one time point is assumed to represent the whole posterior distribution, meaning one sample would encode for the perceptual uncertainty over the stimulus orientation. But on first inspection of draws from our latent space $\mathbf{y}$, the formation of a spatial code is abstract. We proceed with a pragmatic approach by drawing multiple samples from the distribution, $p(\mathbf{y}|\mathbf{x})$, and then averaging these samples to create an instantaneous population code for the high dimensional posterior. This can be decoded under the assumption that samples of $\mathbf{y}$ are in fact noisy estimates of the stimulus and averaging removes the noise (here, noise is seen as a nuisance variable that corrupts uncertainty encoding - training a decoder in this manner allows us to prove or disprove the importance of noise in encoding uncertainty). Specifically:

1. For each stimulus $\mathbf{x}$, draw N_{agg} independent samples from $p(\mathbf{y}|\mathbf{x})$ (like in the temporal scheme).
2. Compute the average: $\bar{\mathbf{y}} = \frac{1}{N_{\text{agg}}} \sum_{i=1}^{N_{\text{agg}}} \mathbf{y}^{(i)}$. We will use this $\bar{\mathbf{y}}$ as the population representation at a single time.

We call N_{agg} the number of temporal samples that are aggregated before being inputted into the spatial neural decoder. Of course if $N_{\text{agg}} = 1$, given the construction of the temporal encoding we have a degenerate case which conveys minimal uncertainty information. In the experimental setting we are limited to 3 temporal samples (Effective Sample Size = 3), hence we choose $N_{\text{agg}} = 3$ to parallel the temporal decoding case.

Thus, for each stimulus $\mathbf{x}$, (by trivial adaptation of the temporal neural data generator through averaging operations described) we produce one vector $\bar{\mathbf{y}}$. This vector is then used as input to a **spatial decoder network** which is trained to output the distribution $p(\theta|\mathbf{x})$. In the temporal decoder case the interpretation of uncertainty came from the sample-to-sample variability (Section 4.2) - for example if the posterior $p(\mathbf{y}|\mathbf{x})$ is broad (high uncertainty), then independent draws $\mathbf{y}^{(i)}$ will vary a lot. But our spatial decoder in a single trial one only sees one $\bar{\mathbf{y}}$, from which it must decode uncertainty. An interpretation in our approach is that the magnitude or pattern of $\bar{\mathbf{y}}$ might systematically depend on uncertainty. For instance, consider orientation coding: if uncertain (for example in low contrast settings), $\bar{\mathbf{y}}$ might tend to be closer to zero or have a more diffuse pattern because in some samples one subset of orientation features dominate and in others another subset of orientation features dominate- averaging them could cancel out some components leading to a broader posterior being encoded. Similarly, in higher contrast settings (where activations are inherently more certain in both schemes), the aggregated activation vector $\bar{\mathbf{y}}$ will have strong components in certain directions as individual samples that contribute to $\bar{\mathbf{y}}$ align.

4.4 Decoder Training Objective

For both the temporal and spatial case, we want the decoder's predicted posterior distribution over orientation $\hat{p}(\theta|\mathbf{x})$ to match the true posterior $p(\theta|\mathbf{x})$ from the ideal observer model. Hence a natural choice of loss is the Kullback-Leibler (KL) divergence. We define the loss for a single stimulus $\mathbf{x}$ as:

$$\mathcal{L}(\mathbf{x}) = D_{\mathrm{KL}}\Big(p(\theta|\mathbf{x}) \,\Big|\, \hat{p}(\theta|\mathbf{x})\Big) = \sum_{\theta} p(\theta|\mathbf{x}) \log \frac{p(\theta|\mathbf{x})}{\hat{p}(\theta|\mathbf{x})} \tag{35}$$

Minimising this over the training data will encourage $\hat{p}$ to be close to p and since $p(\theta|\mathbf{x})$ is a fixed target given by our model, this is equivalent (up to an additive constant) to minimising the cross-entropy $-\sum_{\theta} p(\theta|\mathbf{x}) \log \hat{p}(\theta|\mathbf{x})$. In training we are able to generate a large training set of (theoretically infinite, limited only by the range of parameters of contrast and aperture, as well as the use of discrete orientations) stimuli $\mathbf{x}$. As well as training with experimental gratings, it was also possible to train based on generation with the generative model and sampling for z from the gamma prior and $\mathbf{y}$ from the conditional prior, then constructing the stimuli using Equation (5) - this also provided good test results. For each generated image we compute the target perceptual posterior $p(\theta|\mathbf{x})$ using our Bayesian model - this can be formed either using Equation (17) or Equation (33), depending on whether we are considering the decision variable or not. We also generate the temporal synthetic neural data (again using Algorithm 1 or Algorithm 2) and then either use the sequential readout to train the temporal neural decoder (where we aggregate the posterior predictions) or average the samples and then feed it into the spatial decoder. We proceed to train both neural networks by minimising the cross entropy loss across the image dataset by using the ideal observer posterior as the supervision signal in both cases (see Appendix A.2 for how the experimental images were created).

5 Results and Discussion

Having established a computational framework and trained our decoders, we first verify that our Bayesian ideal observer reveals expected behaviours, validating our use of the ideal observer posteriors as ground truth supervision signals. We then proceed to the central comparison of the neural decoding schemes.

5.1 Differential Modulation of Perceptual and Decision Uncertainty

To establish consistency with Koblinger et al.'s experiments and provide empirical support for using the ideal observer as a ground truth generator, we perform analysis to quantify how uncertainty in the posterior distributions varied with controlled manipulations in the different stimulus parameters: contrast, aperture and orientation. For the nuisance variables, a comprehensive range was selected to reflect the settings that were used in

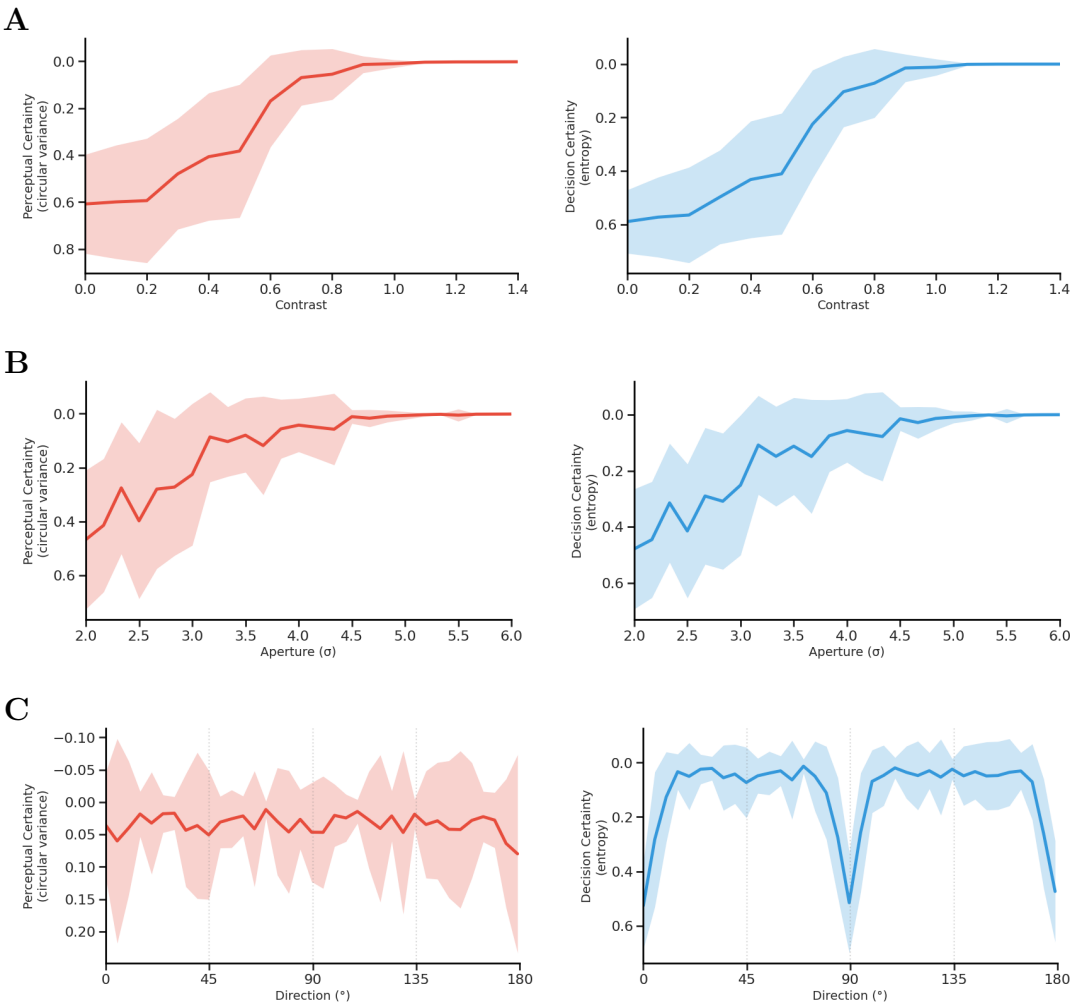


Figure 9: Dissociation between perceptual and decision uncertainty under stimulus manipulations: (A) Contrast modulation and **(B)** Aperture size reveals parallel increases in both perceptual and decision certainty. **(C)** Orientation modulation demonstrates invariant perceptual certainty while decision certainty decreases (increased uncertainty) at the decision boundary. Solid lines show means across 40 trials; shaded regions indicate ±1S.D.

other analyses, and for orientation, thirty-six orientations spanning the full 180° range were tested at at 5° intervals. For each parameter combination, we generated a grating stimuli (as per Appendix A.2) adding observation noise according to the generative model

Perceptual uncertainty was quantified through circular variance of the orientation posterior distribution $p(\theta|\mathbf{x})$, where positive values up to 1 indicate increasing uncertainty and 0 indicates certainty. Decision uncertainty was measured using the Shannon entropy of the binary decision posterior $p(d|\mathbf{x})$. Figure 9 presents the results with axes inverted to display certainty rather than uncertainty, facilitating intuitive interpretation where higher values indicate greater confidence. First, contrast modulation (Figure 9 **A**) reveals parallel increases in both perceptual and decision certainty, with sigmoid-like transitions occurring around 0.5-0.8 contrast. Second, aperture size (Figure 9 **B**) produces modest but consistent improvements in both uncertainty types, with diminishing returns beyond $\sigma = 4$. This plateau suggests that sufficient spatial information for reliable inference is

available within moderate aperture sizes.

The most significant result is the orientation modulation in Figure 9 **C**, which demonstrates a clear dissociation between the two uncertainty types. While perceptual certainty remains invariant across all orientations, indicating the model's ability to estimate any orientation with equal precision, decision certainty exhibits a dramatic drop at 90° (coinciding with the categorical decision boundary). These results suggest a fundamental computational principle: the brain maintains high-fidelity sensory representations while allowing decision uncertainty to reflect task demands.

This dissociation between perceptual and decision uncertainty raises a critical question: *can neural systems learn to decode these distinct forms of uncertainty from neural activity?* By training decoders to recover both perceptual posteriors $p(\theta|\mathbf{x})$ and decision posteriors $p(d|\mathbf{x})$ from latent neural representations, we can assess whether the computational dissociation we observe at the behavioural level is reflected in the learnability and decodability of these distinct uncertainty signals.

5.2 Network Implementation Following Koblinger et al.

Our initial approach aimed to directly replicate the neural decoder architecture used by Koblinger et al. (2024), ensuring consistency between the experimental paradigm and this computational analysis. The original study analysed behavioural and neural data from mouse subjects across approximately 300 trials per animal. Their training protocol used stochastic gradient descent (SGD) from 5 random initialisations, training for 40 epochs with a 1:1 train-test split. This relatively modest data regime - effectively using only 150 trials for training per subject - necessitated a correspondingly constrained neural network architecture to avoid overfitting. To maintain experimental validity and enable direct comparisons across our decoding scenarios, we required identical network architectures across all conditions. This constraint was crucial as any architectural differences between decoders could impair our ability to interpret performance differences or attribute them to the encoding scheme rather than to architectural advantages.

We began with the highest complexity mapping of the latent samples to the orientation posterior. Our original 248-dimensional latent feature input $\mathbf{y}$ was randomly subsampled to reduce the input dimension to 60. The neural network had one hidden layer of 10 neurons, and outputted probabilities over orientation space in discretised bins of 10° (hence a total output dimension of 18 to cover all 180°). Successful training of this network would allow us to proceed with this architecture on all other mappings, from which we could do a fair analysis.

Convergence Failure: Despite extensive hyperparameter exploration and multiple training attempts (with five random initialisations), when trained on KL divergence loss (between target and prediction), the replicated architecture consistently failed to converge to meaningful solutions. We tested the network on four images of varying parameters (Figure 10), and as seen in Figure 11 the network produced poor posterior predictions regardless of input stimuli. While Koblinger et al. successfully trained their decoders on

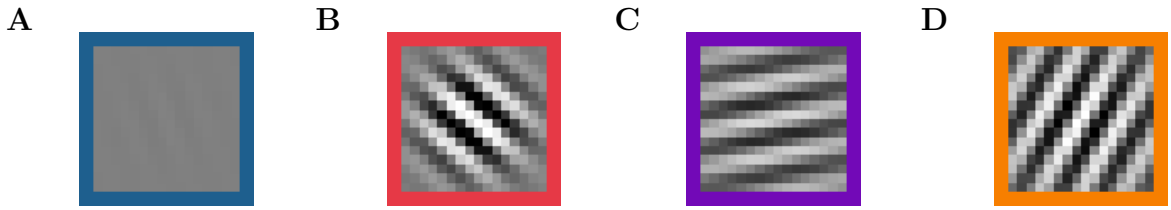


Figure 10: Representative test stimuli demonstrating systematic manipulation of nuisance variables for neural decoder evaluation. **(A)** Low contrast, low aperture: minimal visual information with barely perceptible grating structure. **(B)** High contrast, low aperture: strong orientation signal but spatially restricted viewing window. **(C)** Low contrast, high aperture: weak orientation signal across extended spatial region. **(D)** High contrast, high aperture: strong orientation signal with full spatial extent. These stimuli span the range of uncertainty conditions, from maximum perceptual uncertainty (A) where both contrast and spatial information are limited, to minimum uncertainty (D) where both cues strongly support orientation discrimination. Border colors (blue, red, purple, orange) correspond to the decoder prediction traces shown in subsequent results figures, providing visual correspondence between input stimuli and their decoded posterior distributions.

biological data, our synthetic data, generated from the full posterior distribution $p(\mathbf{y}|\mathbf{x})$, potentially contains more complex statistical structure than the biological recordings.

These results presented a methodological impasse with respect to further usage of this architecture and training protocol. Given that this temporal perceptual decoder failed to converge, it was likely that other decoder variants (spatial perceptual, temporal decision, and spatial decision) would likely face similar limitations. Proceeding with this architecture would also compromise our ability to make meaningful comparisons between the encoding schemes. These results motivated our transition to more expressive decoder architectures. We adopted feed forward networks with two hidden layers of 512 units each, employing ReLU activations. These networks were trained on 500,000 images over 5 epochs, and we could make use of our full 248 dimensional latent space as the input (no sub-sampling). Details of the training can be found in Section 4.4. In the following section we carry out the comparisons between the different decoding schemes, and our use of flexible, high-capacity decoders ensure that our conclusions about temporal versus spatial coding do not depend on specific architectural choices.

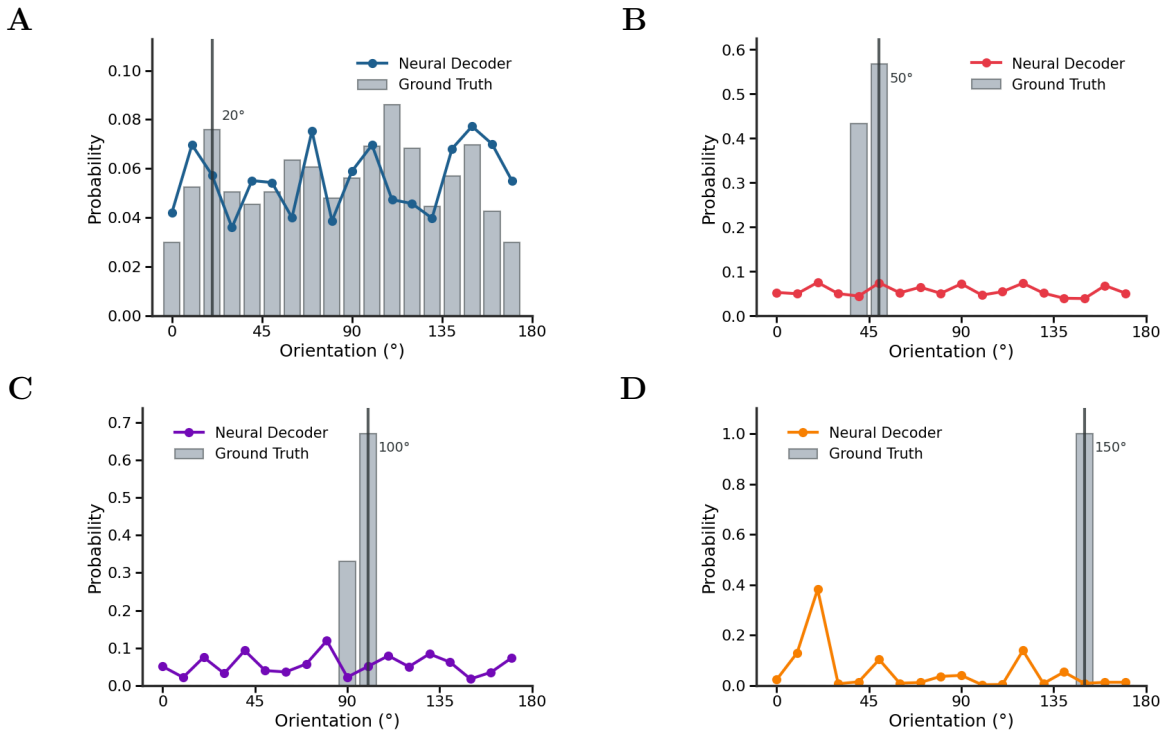


Figure 11: Low-capacity neural decoder fails to recover orientation posteriors for test stimuli from Figure 10. Decoder predictions (colored lines) remain near-uniform (0.05-0.1 probability) across all orientations regardless of ground truth posteriors (gray bars) or stimulus conditions. True orientations: **(A)** 20°, **(B)** 50°, **(C)** 100°, **(D)** 150°. This systematic failure to capture any peaked structure indicates the network learned no meaningful mapping from neural activity to orientation uncertainty, motivating the transition to higher-capacity architectures.

5.3 Comparative Analysis of High-Capacity Neural Decoders

All neural decoders learned the mapping from the latent neural activity and as shown, visual inspection alone reveals very similar outputs - convergence in both cases (temporal and spatial decoders, which share the same architecture for perception and decision) make it difficult to discern which encoding scheme actually captures the uncertainty more faithfully. Given the high capacity of these decoders, both architectures do indeed have sufficient expressiveness to approximate the target distributions well. Hence the question remains of which encoding scheme better captures uncertainty representation in V1, and clearly small but systematic changes are imperceptible by visual inspection so we look into computing KL divergences (normalised by maximum entropy of the output) between the predictions and ground truths, and establish whether these differences are statistically meaningful or just random variation inherited from the training process.

To test the competing coding hypotheses, we use paired permutation testing, which is well suited to our problem as we can compare the observed differences against a null distribution generated through random permutation, allowing us to quantify exactly how unlikely our results would be if the two decoding schemes were truly equivalent. We conduct this analysis across all four combinations of data source (Perceptual vs Decision Bayesian ideal observer) and decoder target (perceptual vs decision uncertainty) as shown

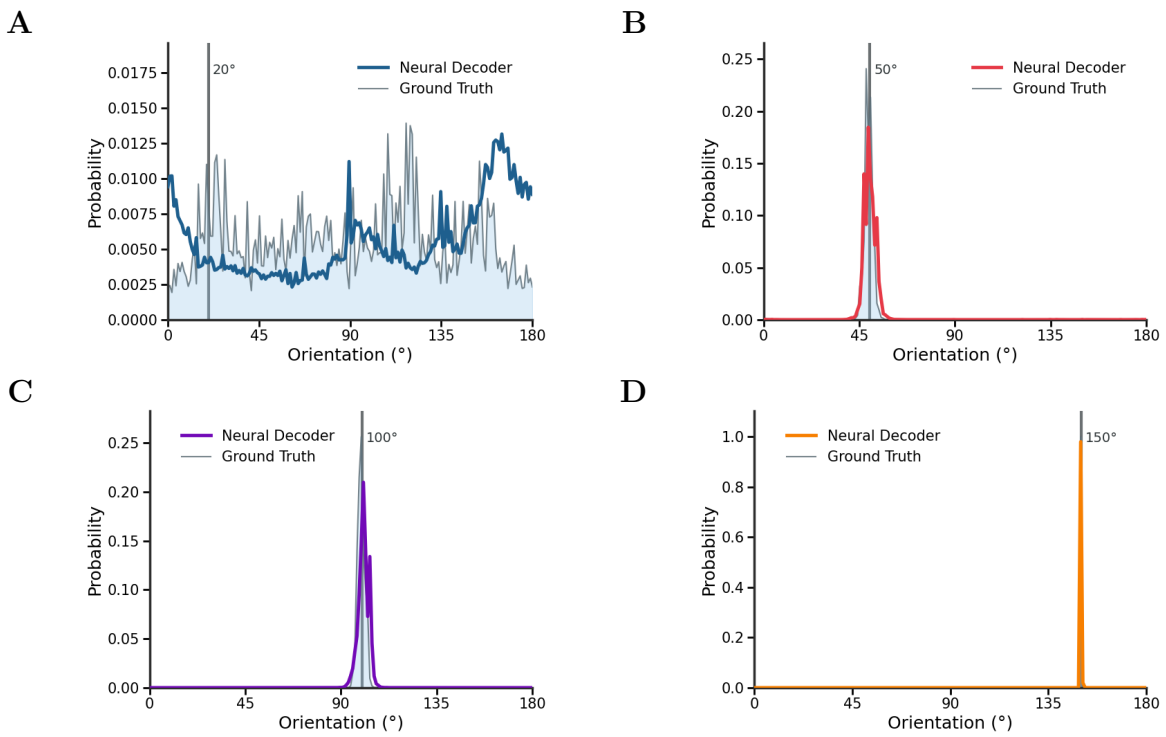


Figure 12: High-capacity temporal perceptual decoder successfully recovers orientation posterior distributions across all test stimuli from Figure 10. Decoder predictions (colored lines) closely align with Bayesian ideal observer posteriors (light blue) across all uncertainty conditions: **(A)** broad posterior under high uncertainty, **(B-D)** progressively sharper posteriors as stimulus information increases. The successful recovery of both broad and peaked distributions demonstrates effective learning of the temporal encoding-to-uncertainty mapping.

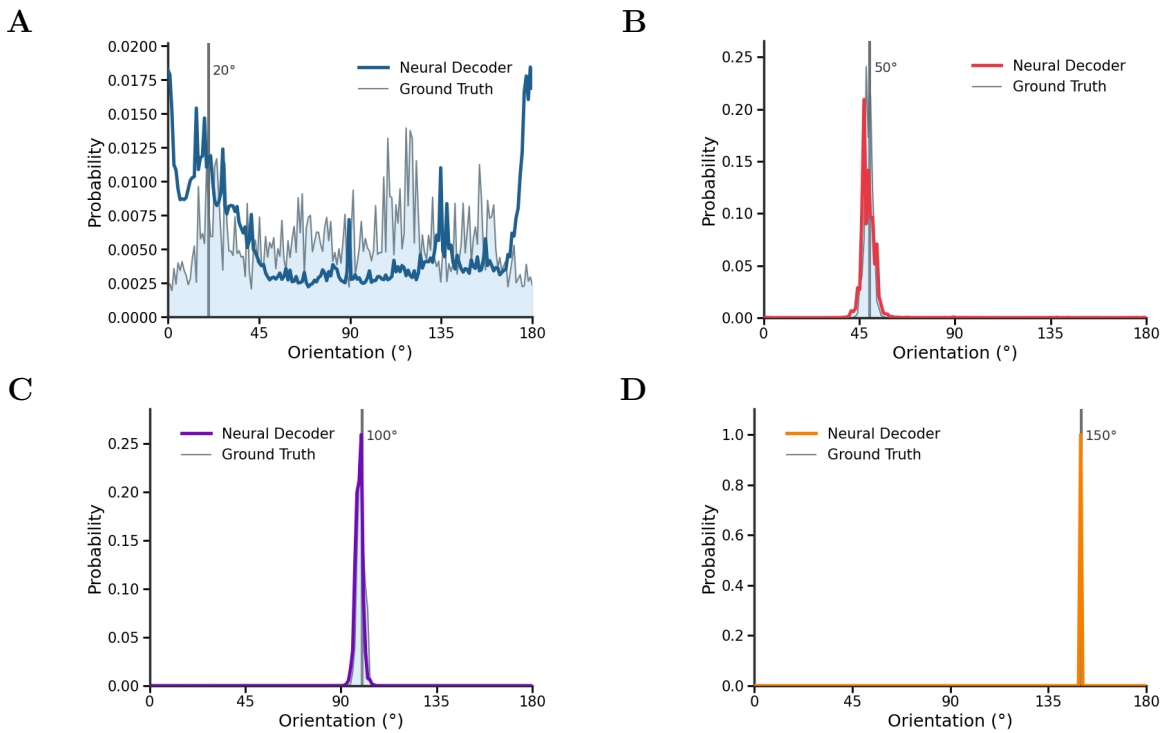


Figure 13: High-capacity spatial perceptual decoder performance on identical test stimuli. Visual inspection reveals comparable performance to the temporal decoder (Figure 12), with accurate recovery of both broad uncertainty distributions **(A)** and sharp, peaked posteriors **(B-D)**. The similar visual performance between temporal and spatial decoders necessitates quantitative analysis via KL divergence to distinguish their relative effectiveness.

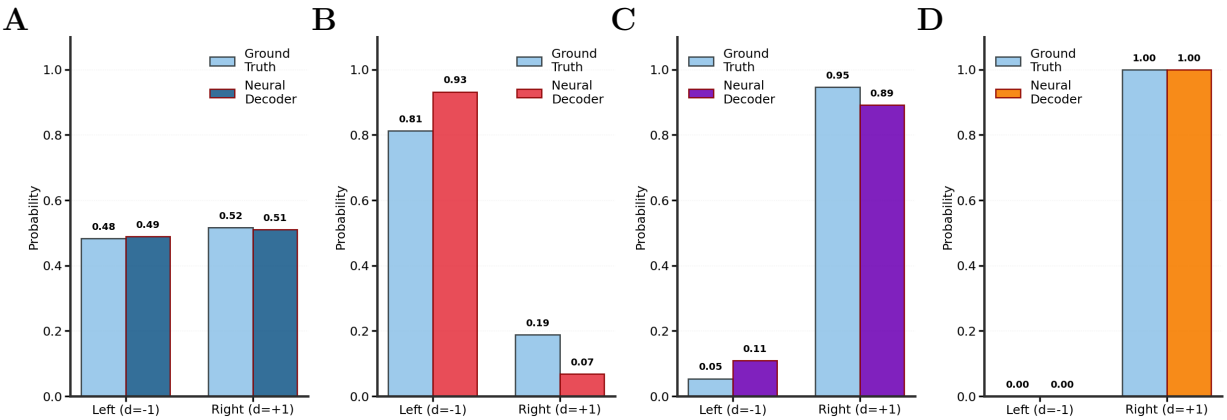


Figure 14: High-capacity temporal decision decoder accurately recovers binary decision posteriors for test stimuli from Figure 10. Ground truth (light blue bars) and decoder predictions (coloured bars matching test stimuli borders) show close correspondence across all conditions: **(A)** balanced probabilities for ambiguous 20° stimulus, **(B)** strong left preference for 50°, **(C)** strong right preference for 100°, **(D)** maximum right certainty for 150°. The decoder successfully learns to map temporal samples to categorical decision probabilities.

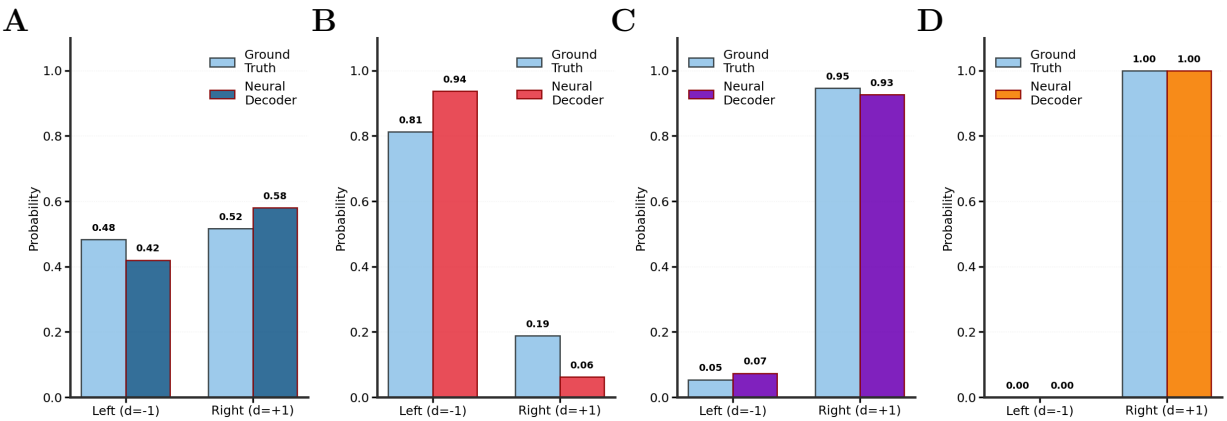


Figure 15: High-capacity spatial decision decoder performance on identical test stimuli. As with the temporal decoder (Figure 14), spatial decoding achieves excellent correspondence between ground truth and predictions across all decision conditions. The comparable visual performance between temporal and spatial decision decoders, combined with the simpler binary output space, suggests that instantaneous population codes may be sufficient for categorical decision representation.

in Figure 16, and we yield two key findings.

Temporal outperforms spatial for perceptual decoding: As shown in Figure 16 **A** and **B**, we find that our temporal decoding outperforms spatial decoding for perceptual uncertainty - this provides independent validation of the results of Koblinger et al. (2024)[10] under our framework. Across both perceptual and decision Bayesian ideal observer (BIO) data sources, temporal decoders achieved mean normalised KL divergences

of 0.103 ± 0.094 and 0.101 ± 0.087 respectively, compared to spatial decoders' 0.128 ± 0.165 and 0.123 ± 0.162. The consistency across BIO types - whether neural samples $\mathbf{y}$ were generated through perceptual inference model or through decision-constrained inference model - demonstrates that temporal coding's advantage is robust to the generative process. This aligns precisely with Koblinger et al.'s experimental findings in mouse V1, where decoders integrating information across multiple time bins consistently outperformed those using instantaneous activity patterns. Our results are statistically significant ($p < 0.001$) and strongly supports the neural sampling hypothesis: V1 represents perceptual uncertainty through stochastic dynamics rather than static population codes, with each of our three temporal samples contributing unique information about the posterior distribution.

Decision variable decoding should be at least as good as perceptual variable decoding: Results from our analysis reveals a systematic superiority of decision decoders over perceptual decoders. The hierarchical model in Figure 8 follows a causal structure where the decision variable d constrains θ to either left $(0 - 90°)$ or right $(90 - 180°)$ categories, θ determines the distribution of the latent features $\mathbf{y}$ through orientation conditioned priors (see Section 3.2) and finally $\mathbf{y}$ and z generates the observed image $\mathbf{x}$. Given this hierarchical structure, a sufficiently flexible decoder should find it easier to extract the coarse binary decision information (which only requires determining which half of orientation space generated $\mathbf{y}$) than to recover the full orientation posterior with 180 output values $(0 - 180°)$. Our high capacity MLPs are indeed sufficiently expressive, with two hidden layers of 512 neurons and over 450,000 images seen in training. Decision decoders achieve significantly lower KL divergences compared to perceptual decoders as seen in Figure 16. As per the statistically significant results of our permutation tests, the coarseness of the categorical variables could also be proposed to explain why spatial decoding outperforms temporal decoding for decision uncertainty. Since decision posteriors are inherently simpler, a single instantaneous population pattern may encode sufficient information for accurate decoding. The averaging process that creates our proxy spatial code effectively integrates evidence across time into a stable population vector, which might capture all relevant information about binary choice probability. In contrast, the temporal sampling approach, which is provably superior at encoding complex perceptual distributions, may be unnecessary in decision decoding. The sample-to-sample variability that enriches perceptual posterior representation might potentially corrupt the straightforward categorical posterior. However, this is a result that should be explored more in future work, with special consideration given to the effectiveness of feed forward networks as neural decoders.

While our permutation tests establish the statistical superiority of temporal coding for perceptual uncertainty, it remains to be understood whether a temporal decoder genuinely benefits from integrating multiple samples, or if similar performance can be achieved with single samples.

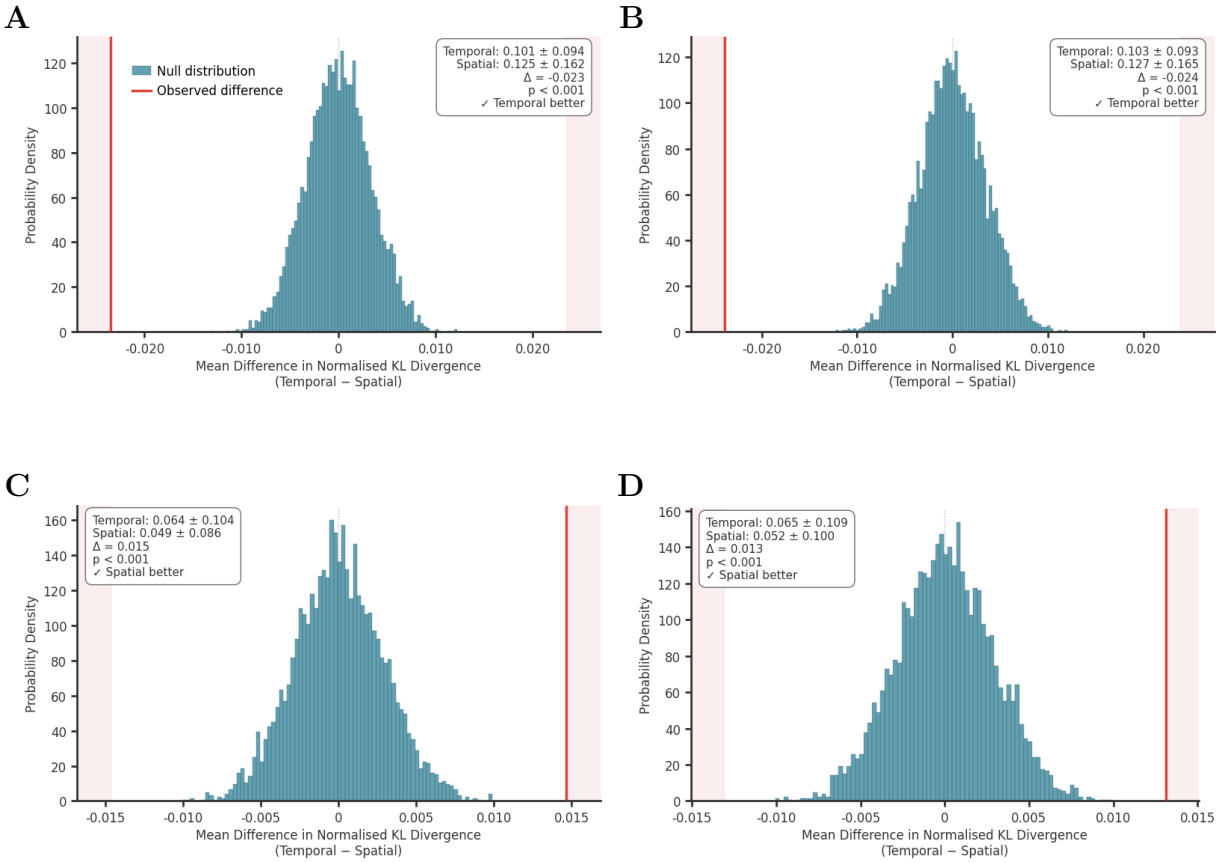


Figure 16: Paired permutation tests reveal task-dependent optimal neural decoding schemes for uncertainty representation. Histograms show null distributions of mean differences in normalised KL divergence between temporal and spatial decoders across 1,260 test images, generated from 5,000 permutations. Red vertical lines indicate observed differences. Shaded regions denote two-tailed rejection regions ($\alpha = 0.05$). Statistical summaries in insets show mean $\pm$ standard deviation of normalised KL divergences and which decoding scheme performs better. **(A)** Perceptual Bayesian Ideal Observer (BIO) samples decoded by perceptual MLPs: temporal decoding outperforms spatial ($\Delta = -0.025$). **(B)** Decision BIO samples decoded by perceptual MLPs: temporal decoding outperforms spatial ($\Delta = -0.022$). **(C)** Perceptual BIO samples decoded by decision MLPs: spatial decoding outperforms temporal ($\Delta = 0.016$). **(D)** Decision BIO samples decoded by decision MLPs: spatial decoding outperforms temporal ($\Delta = 0.013$). All comparisons yield $p < 0.001$, indicating highly significant differences between decoding schemes.

5.4 The Critical Role of Sample Integration in Temporal Decoding

To directly test whether temporal coding benefits from sample integration, two separate neural decoders were trained, either with access to single samples or to three separate samples from the posterior $p(\mathbf{y}|\mathbf{x})$. If it was the case that neural activity just represents noisy point estimates of information rather than samples from a posterior, we would expect only minimal improvement from the three-sample decoder compared to the single sample decoder. Conversely, if (as the temporal coding hypothesis posits) each sample carries unique uncertainty information, the decoder with access to multiple samples should perform better at posterior recovery.

Figure 17 presents the distribution of KL divergences across a fixed test set (1260 images of varying parameters) for both neural decoder predictions with respect to the ground truth posterior over orientations (perceptual posteriors $p(\theta|\mathbf{x})$). The single-sample decoder (blue) exhibits a broader distribution and a median KL of 0.84, compared to the three-sample decoder which has a median KL of 0.60 when averaged over the 1260 predictions - this showed that the temporal integration indeed aided perceptual posterior decoding accuracy.

To probe which aspects of posterior recovery benefit most from temporal integration, we proceeded to decompose the decoder posteriors into their circular mean (representing the perceived orientation information) and their circular variance (representing uncertainty information), as shown in Figure 18.

If neural responses were simply noisy measurements of a fixed quantity, additional samples would primarily improve mean estimation through noise reduction. Instead, we observe that temporal integration specifically enhances uncertainty quantification. As shown in Figure 18 **A** and **B**, both single-sample and three-sample decoders accurately recover the circular mean, achieving high correlations of $r = 0.938$ and $r = 0.983$ respectively with the Bayesian ideal observer circular mean - this high performance shows that individual samples from $p(\mathbf{y}|\mathbf{x})$ already contains significant information about the true orientation value. The advantage of temporal sampling emerges in uncertainty estimation (Figure 18 **C** and **D**). The single-sample decoder struggles to accurately recover circular variance ($r = 0.736$), with substantial scatter away from the diagonal (points along the diagonal show correspondence with the ideal observer). By contrast, the three-sample decoder achieves significantly improved variance estimation ($r = 0.928$), suggesting that even with the minimal sample size imposed by experimental constraints, meaningful uncertainty information can be recovered. The improvements in posterior recovery, which is modest for circular mean ($\Delta r = 0.045$), but substantial for circular variance ($\Delta r = 0.192$), provides compelling evidence for the sampling based temporal coding hypothesis.

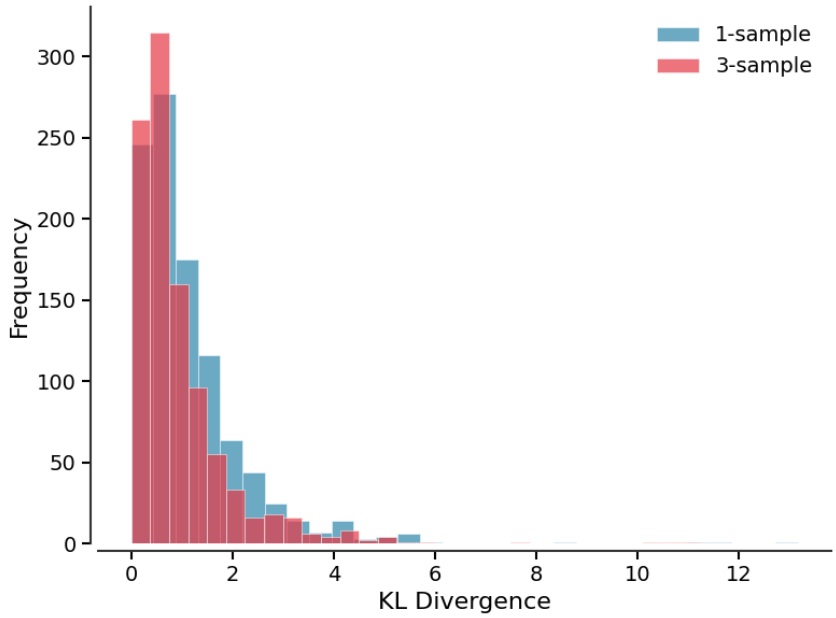


Figure 17: Distribution of KL divergences between temporal perceptual decoder predictions and ground truth posteriors for single-sample versus three-sample temporal decoder. Blue histogram shows performance when decoder has access to only one sample from $p(\mathbf{y}|\mathbf{x})$, while red histogram shows performance with three samples. The leftward shift of the three-sample distribution indicates systematic improvement in posterior recovery, with median KL divergence reducing. This demonstrates that temporal integration of multiple samples provides meaningful information beyond what is available in individual samples.

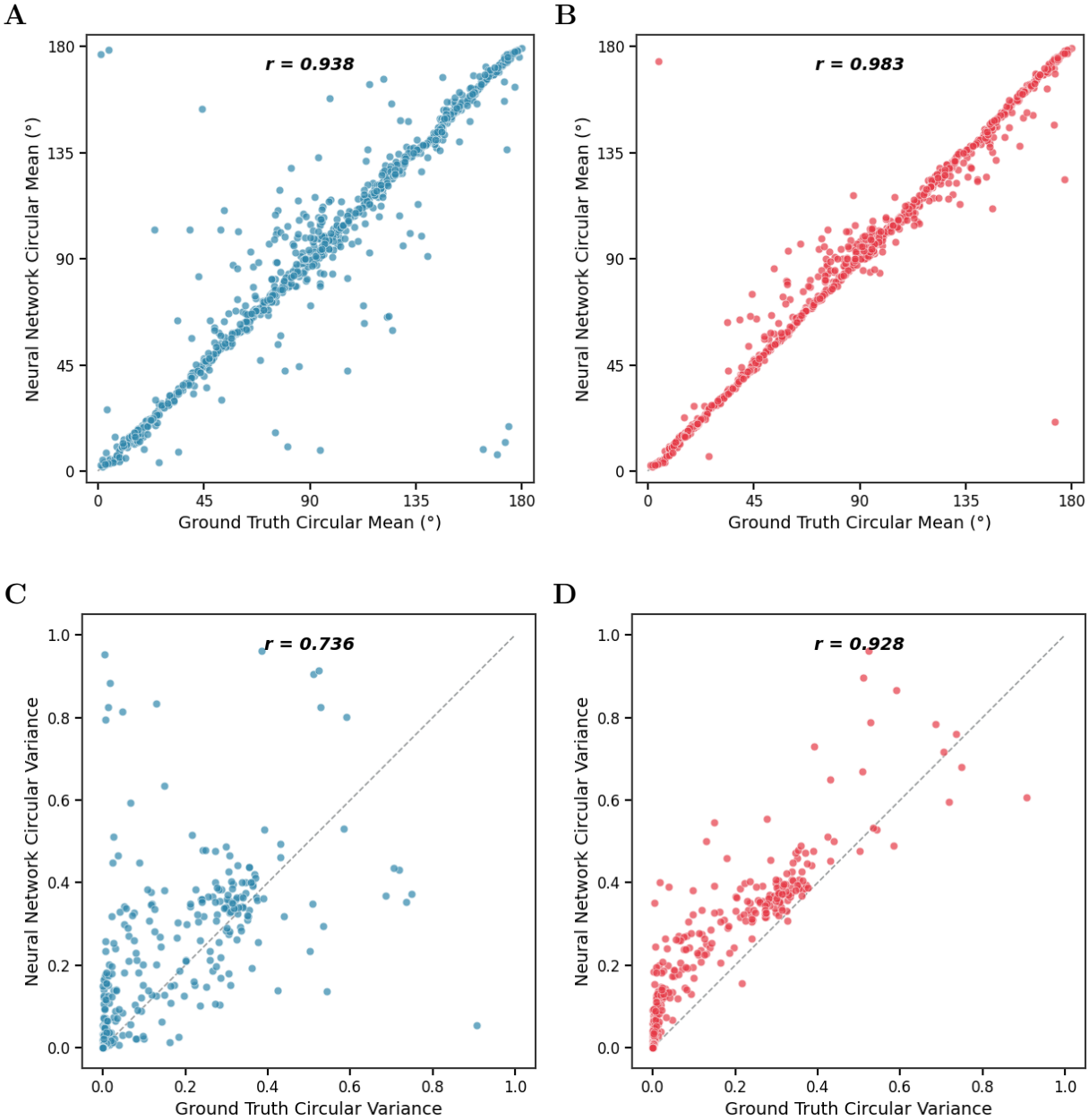


Figure 18: Temporal sample integration improves both mean and variance estimation in orientation posteriors. **(A-B)** Circular mean: correlation between ground truth and decoded circular means for single-sample (A, blue, $r = 0.938$) and three-sample (B, red, $r = 0.983$) temporal decoding. **(C-D)** Circular variance: correlation between ground truth and decoded circular variance for single-sample (C, blue, $r = 0.736$) and three-sample (D, red, $r = 0.928$) decoding. Dashed lines indicate perfect correspondence.

5.5 Potential Extensions to Spatial Encoding Schemes: Preliminary DDC Implementation

The spatial coding hypothesis merits further investigation through more sophisticated encoding schemes. The aggregation approach used so far (where we average temporal samples to serve as a proxy for population-level uncertainty representation during decoding) represents a simplified instantiation of spatial coding - but our ground truth encoding scheme was nonetheless a sampling based temporal one in these cases. To extend to a more rigorous implementation, we implemented a Distributed Distributional Code (DDC) framework, which provides a more theoretically grounded approach of representing posterior distributions through population activity.

The DDC framework, as introduced by Vértes and Sahani [8] uses a large set of nonlinear encoding functions to represent a posterior distribution (see Section 2.2). Specifically, the posterior distribution $p(\mathbf{y}|\mathbf{x})$ is represented by a vector of expected values:

$$r^{(i)}(\mathbf{x}) = \mathbb{E}_{p(\mathbf{y}|\mathbf{x})}\left[\sigma(\mathbf{w}^{(i)T}\mathbf{y} + b^{(i)})\right] \quad \text{for } i = 1, \ldots, K \tag{36}$$

where $\mathbf{w}^{(i)} \in \mathbb{R}^{D_y}$ are random projection vectors sampled from a standard normal distribution, $b^{(i)}$ are similarly distributed bias terms, and $\sigma(\cdot)$ is the sigmoid nonlinearity. Here we adopt $K = 1024$ to represent our $D_y = 248$ dimensional posterior $p(\mathbf{y}|\mathbf{x})$.

The key computational challenge lies in evaluating these expectations over the complex posterior distribution derived from our Bayesian ideal observer model. Recall from Section 3.3 that our posterior takes the form:

$$p(\mathbf{y}|\mathbf{x}) = \sum_{\theta} p(\theta|\mathbf{x}) \sum_{z} p(\mathbf{y}|z, \theta, \mathbf{x}) p(z|\mathbf{x}, \theta) \Delta z \tag{37}$$

For each component of the mixture, $p(\mathbf{y}|z, \theta, \mathbf{x})$ is Gaussian with parameters derived through our inference procedure:

$$p(\mathbf{y}|z, \theta, \mathbf{x}) = \mathcal{N}(\mathbf{y}; \boldsymbol{\mu}_{z,\theta}, \boldsymbol{\Sigma}_{z,\theta}) \tag{38}$$

where $\boldsymbol{\Sigma}_{z,\theta}$ and $\boldsymbol{\mu}_{z,\theta}$ are computed according to Equation (21) and Equation (22). To compute the DDC expectations, we proceed to develop an analytical quadrature approach that exploits the Gaussian structure within each mixture component. For a given encoding function indexed by i and mixture componet indexed by (θ, z), we first observe that the scalar projection:

$$y_i' = \mathbf{w}^{(i)T}\mathbf{y} + b^{(i)} \tag{39}$$

follows a univariate Gaussian distribution:

$$y_i' \sim \mathcal{N}(m_{i,z,\theta}, v_{i,z,\theta}) \tag{40}$$

where $m_{i,z,\theta} = \mathbf{w}^{(i)T}\boldsymbol{\mu}_{z,\theta} + b^{(i)}$ and $v_{i,z,\theta} = \mathbf{w}^{(i)T}\boldsymbol{\Sigma}_{z,\theta}\mathbf{w}^{(i)}$. The expectation of the sigmoid

function under this Gaussian distribution:

$$\mathbb{E}_{y'_i \sim \mathcal{N}(m,v)}[\sigma(y'i)] = \int_{-\infty}^{\infty} \sigma(y'_i) \frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{(y'_i - m)^2}{2v}\right) dy'_i \tag{41}$$

lacks a closed-form solution but can be accurately approximated through numerical quadrature. An efficient grid based approach exploits the invariance of normalised quadrature weights under linear transformations. Specifically, for any Gaussian $\mathcal{N}(m, v)$, we can write:

$$\mathcal{N}(y'; m, v) = \frac{1}{\sqrt{v}} \mathcal{N}\left(\frac{y' - m}{\sqrt{v}}; 0, 1\right) \tag{42}$$

When constructing normalised quadrature weights, the scaling factor $1/\sqrt{v}$ cancels in both numerator and denominator, hence we constructed a standardised quadrature grid for $\mathcal{N}(0, 1)$ spanning $[-4, 4]$, with $\{k_j\}_{j=1}^{N_g}$ indexing grid points and $N_g = 100$ denoting the number of grid points. We then precompute and store the normalised weights $\{w_j\}_{j=1}^{N_g}$. The expectation is computed as:

$$\mathbb{E}[\sigma(y'_i)] \approx \sum_{j=1}^{N_g} w_j \cdot \sigma(y'_{i,j}) \tag{43}$$

The complete DDC expectation for the ith encoding function combines contributions from all mixture components:

$$r^{(i)}(\mathbf{x}) = \sum_{\theta} p(\theta|\mathbf{x}) \sum_{z} p(z|\mathbf{x}, \theta) \Delta z \cdot \mathbb{E}_{y'_i \sim \mathcal{N}(m_{i,z,\theta}, v_{i,z,\theta})}[\sigma(y'_i)] \tag{44}$$

$$= \sum_{\theta} p(\theta|\mathbf{x}) \sum_{z} p(z|\mathbf{x}, \theta) \Delta z \cdot \underbrace{\sum_{j=1}^{N_g} w_j \cdot \sigma(m_{i,z,\theta} + \sqrt{v_{i,z,\theta}} \cdot k_j)}_{\text{using precomputed weights}} \tag{45}$$

To validate our DDC implementation and explore its representational capacity, we computed orientation tuning curves for a subset of the 1024 encoding units by presenting high-contrast gratings. Figure 19 demonstrates that the DDC framework successfully captures orientation-specific information through its nonlinear encoding functions. The preferred orientations, marked by vertical red lines, are distributed across the full orientation range, suggesting that the DDC representation provides comprehensive coverage of the orientation space. Notably, the tuning curves exhibit considerable diversity in their response profiles, including unimodal, bimodal and multimodal curves. This heterogeneity is precisely what enables DDCs to represent arbitrary posterior distributions - different combinations of these diverse basis functions can approximate complex uncertainty patterns that simple Gaussian tuning curves (as used in traditional spatial coding schemes) cannot capture. Here we must make a clear distinction: unlike the orientation tuning curves shown previously for individual basis function in the latent feature vector $\mathbf{y}$ (as per Figure 5), these DDC tuning curves represent how different dimensions of the uncertainty encoding vary with stimulus orientation. Each dimension captures a different nonlinear projection of the full posterior distribution $p(\mathbf{y}|\mathbf{x})$, rather than representing the activity of a single latent feature y_i.

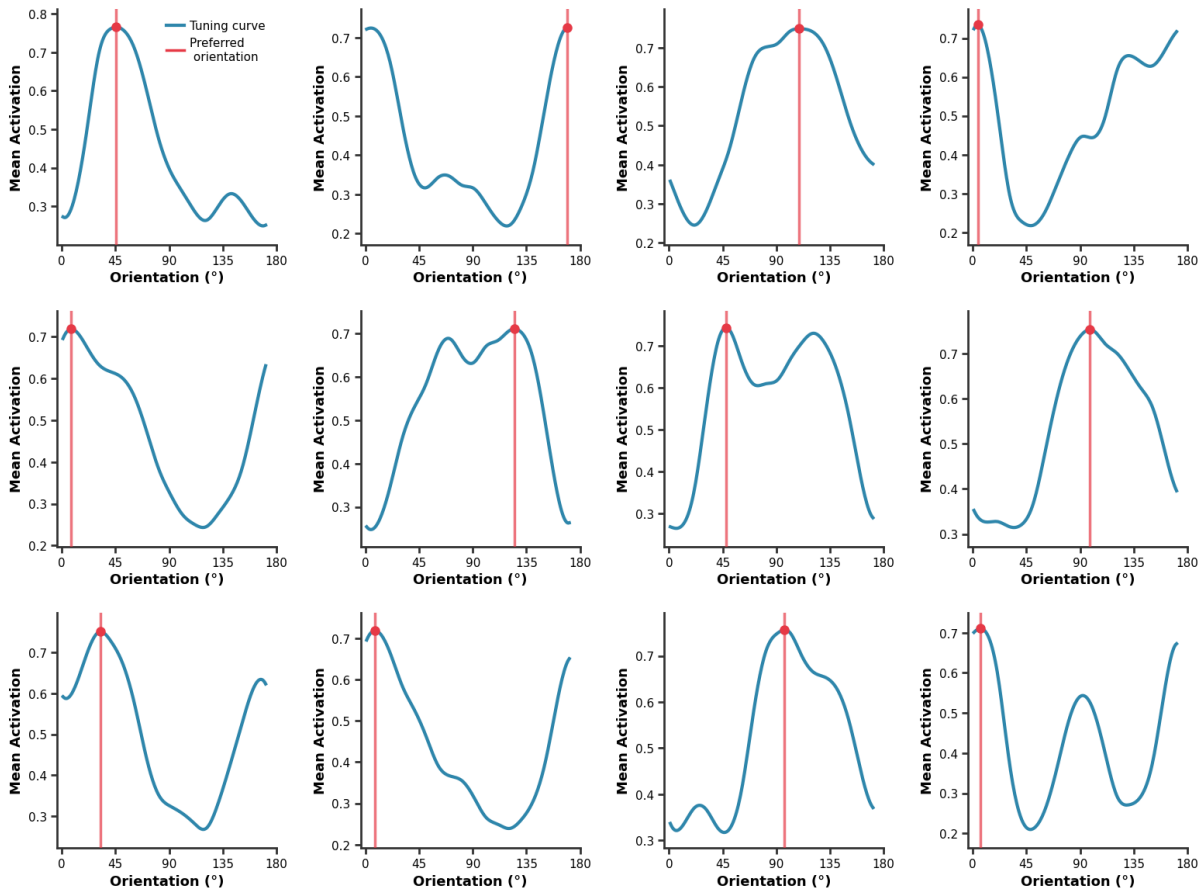


Figure 19: Diverse orientation tuning profiles emerge from DDC encoding functions: the panels show the tuning curves (blue) of the top 12 DDC dimensions with strongest orientation selectivity, computed by presenting oriented gratings ($0 - 180°$) at high contrast and measuring the expected activation $r^{(i)}(\mathbf{x})$. Red vertical lines indicate each unit's preferred orientation.

While these results confirm that DDC representations preserve orientation information in a distributed population code, time constraints prevent the complete development of the encoding-decoding pipeline necessary for quantitative comparison with temporal coding schemes. Specifically, implementing neural decoders that can invert the DDC representation to recover posterior distributions would incur a huge computational overhead (1024 dimensions $\times$ 180 orientations $\times$ 100 grid points per stimulus), deeming large-scale decoder training infeasible within the project timeline. These preliminary results nonetheless establish the computational framework and demonstrate DDC's representational richness, providing essential groundwork for future studies to conduct the full comparative analysis.

6 Conclusions

6.1 Summary of Findings and Contributions

This thesis aimed to advance scientific understanding of how the primary visual cortex encodes uncertainty. To achieve this, it built towards providing robust computational models of encoding schemes and decoding strategies using neural networks, that would be part of an analysis pipeline. Indeed, through developing a Bayesian ideal observer framework to construct the encoding (temporal neural data generator) and ground truth, followed by systematic testing and comparison of decoding strategies on synthetic data, the work has revealed differences in how well temporal versus spatial decoding can extract perceptual and decision uncertainty.

The primary contribution of the work that has been done over the last year lies in establishing a rigorous computational framework for neural coding hypotheses. Building on the Orbán et al. (2016) Gaussian Scale Mixture model, we introduced orientation conditioned priors - this mathematical modification, taking us from unstructured general priors to these empirically conditioned priors was the first step that allowed us to extend the generative model. First it was extended to orientation, from which we could obtain perceptual posteriors, and then the framework was further enriched by incorporating decision variables. This allowed us to create ground truth posterior distributions to quantify perceptual uncertainty and decision uncertainty (from $p(\theta|\mathbf{x})$ and $p(d|\mathbf{x})$ respectively). As a result, we were also able to investigate how uncertainty representations might transform from continuous perceptual quantities to discrete behavioural outputs. The results in Section 5.1 gave our mathematical framework critical validation and allowed us to proceed with using the ideal observers to train and compare neural decoders.

For perceptual uncertainty (which we quantified through the posterior distribution over stimulus orientation), our findings on synthetic data showed agreement with the neural sampling hypothesis and the results obtained by Koblinger et al.(2024). The permutation tests in Section 5.3 showed that the temporal decoders (which preserve sample-by-sample information from the encoding process), consistently outperformed spatial decoders that operated on averaged activity vectors $\bar{\mathbf{y}}$. With temporal decoding superiority established in the perceptual case, a further analysis was done in Section 5.4 that revealed the importance of multiple temporal samples in variance estimation of the posterior. While single-sample (not averaged) and three-sample temporal decoders had similar success in posterior mean estimates, the three-sample temporal decoder captured the uncertainty about those estimates. Obtaining a correlation coefficient with the ideal observer of $r = 0.928$ was also significant here as it provided proof that although our experimental setup was limited by three temporal samples (ESS = 3), this was sufficient for our computational framework. Overall, the computational results in the temporal decoding of perceptual uncertainty support the view that trial-to-trial variability might carry essential information in biological systems.

The unexpected superiority of spatial decoding for decision uncertainty presents a point that merits further investigation. This result was persistent and statistically significant as shown by the permutation tests in Section 5.3, and it challenges the assumption that

temporal decoding should universally outperform spatial decoding given temporal encoding (as is the case we have). One explanation for this result is that for coarser binary decisions, averaging might perform appropriate evidence integration. Another is that our neural network decoders might have inductive biases towards such averaging operations for easier learning objectives such as this mapping from latent space to binary categories. These are mere hypotheses and without further study or biological validation, no definitive claims can be made about whether the results we obtained with respect to decision decoding reflect a principle of neural computation or an artifact of our methodology.

6.2 Limitations and Future Directions

The critical limitation of the work that prevents it from being a complete encoding/decoding analysis framework is that all results derive from synthetic data with known temporal encoding. The true biological coding scheme of V1 remains unknown and awaits empirical determination through experimental results, but our framework nonetheless provides tools for testing the coding hypotheses on real data. The spatial decoding tested here is an oversimplification, and the preliminary DDC implementation demonstrates that true spatial encoding, where uncertainty is directly represented in instantaneous population patterns, requires further analysis.

Another point of consideration is the biological faithfulness of our computational encoding framework, which involves various simplifications - for one, our temporal neural data generator takes samples (assumed to be i.i.d.) from a posterior distribution directly. Work by Echeveste et al.(2020)[15] highlighted that real neural populations exhibit more complex dependencies through recurrent interactions and stimulus-modulated dynamics that our simplifying assumptions ignore. Similarly, while the Gaussian Scale Mixture model we adopted from Orbán (2016) successfully captures image statistics, it simplifies the non-linear cortical computations that Echeveste et al. show can arise from biologically-constrained neural circuits. In terms of decoding, as already established following our unexpected result of decision decoding, where spatial decoders performed better than temporal decoders, we must explore how decoder choice can critically influence conclusions about neural coding.

Future work must address these limitations systematically. The immediate priority is completing the DDC implementation to enable comparison between true spatial encoding and temporal encoding. This would work towards a more complete computational framework for testing the coding hypotheses. Although an attempt was made to train neural decoders on the DDC representations, time constraints made this challenging as little opportunity was given to make the appropriate computational optimisations. Here the bottleneck was the use of 1024 encoding functions and the numerical quadrature required for the expectation calculations - a more principled approach may be to start with a smaller number of encoding functions and simultaneously design more efficient training routines and neural decoder architectures.

Another more easily addressed point is to close the gap between the current computational model and the actual experimental set up. This would include establishing better priors over orientation and decision (rather than uniform priors); a method touched on

in Section 3.4 was that of fitting priors to data. Additionally, we must investigate further as to why the neural decoder architecture used by Koblinger et al. yielded poor results.

6.3 Final Thoughts

This work establishes essential tools for future investigations: a robust method for generating ground-truth posteriors, biologically-constrained synthetic data generation, and systematic comparison protocols through expressive neural decoders. This thesis has demonstrated the value of computational frameworks for investigating neural coding and importantly in developing these frameworks, the results obtained have highlighted how our methodological choices (whether it be decoder architecture or training protocols) can influence our results.

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Appendix

A Proofs and Derivations

A.1 Reconstruction Error: Pairwise Similarity

In this section, we quantify the similarity between latent representations corresponding to different orientations by comparing the subspaces spanned by their dominant principal components. For a given orientation θ, we extract the top two principal components from the latent activity and define a corresponding two-dimensional subspace (Plane 1) spanned by the orthonormal vectors $\{\mathbf{u}_1, \mathbf{u}_2\}$. Similarly, for another set of latent samples (or a different condition), we define Plane 2 using its two leading principal components $\{\mathbf{v}_1, \mathbf{v}_2\}$. Our goal is to assess how well the dominant feature of one subspace (specifically, $\mathbf{v}_1$ from Plane 2) can be reconstructed using the basis vectors of the other subspace (i.e. $\mathbf{u}_1$ and $\mathbf{u}_2$ from Plane 1). To this end, we reconstruct $\mathbf{v}_1$ as

$$\widehat{\mathbf{v}}_1 = \alpha_{12}^{(1)}\,\mathbf{u}_1 + \alpha_{12}^{(2)}\,\mathbf{u}_2,$$

where the coefficients $\alpha_{12}^{(1)}$ and $\alpha_{12}^{(2)}$ are determined by minimising the squared reconstruction error

$$e_{12} = \|\mathbf{v}_1 - \widehat{\mathbf{v}}_1\|^2.$$

A smaller error e_{12} indicates that the two latent subspaces are more similar, implying that the latent representations are well-aligned across conditions or orientations. This analysis helps us to understand the structure and consistency of the latent space with respect to the orientation of the stimuli.

Defining the Planes

For a given orientation θ, we can define a plane spanned by its top two principal components:

$$\text{Plane 1: } \{\,\mathbf{u}_1,\, \mathbf{u}_2\}, \quad \text{Plane 2: } \{\,\mathbf{v}_1,\, \mathbf{v}_2\}.$$

We assume these vectors are unit-length and orthogonal within each plane:

$$\mathbf{u}_1^T\mathbf{u}_2 = 0, \quad \mathbf{v}_1^T\mathbf{v}_2 = 0, \quad \|\mathbf{u}_1\| = \|\mathbf{u}_2\| = \|\mathbf{v}_1\| = \|\mathbf{v}_2\| = 1.$$

Reconstruction Error

We want to compare, for instance, $\mathbf{v}_1$ (the first principal component of Plane 2) with its reconstruction in Plane 1. Denoting the reconstructed vector by $\widehat{\mathbf{v}}_1$:

$$\widehat{\mathbf{v}}_1 \;=\; \alpha_{12}^{(1)}\,\mathbf{u}_1 \;+\; \alpha_{12}^{(2)}\,\mathbf{u}_2,$$

where $\alpha_{12}^{(1)}$ and $\alpha_{12}^{(2)}$ are chosen to minimise the squared error

$$e_{12} \;=\; \left\|\mathbf{v}_1 \;-\; \widehat{\mathbf{v}}_1\right\|^2.$$

Closed-Form Solution

By taking derivatives and imposing optimality conditions, we find

$$\alpha_{12}^{(1)} \;=\; \mathbf{u}_1^T\mathbf{v}_1 \quad \text{and} \quad \alpha_{12}^{(2)} \;=\; \mathbf{u}_2^T\mathbf{v}_1.$$

Denoting

$$b_1 \;=\; \mathbf{u}_1^T\mathbf{v}_1 \quad \text{and} \quad b_2 \;=\; \mathbf{u}_2^T\mathbf{v}_1,$$

it follows that

$$\widehat{\mathbf{v}}_1 \;=\; b_1\,\mathbf{u}_1 \;+\; b_2\,\mathbf{u}_2 \quad \text{and} \quad e_{12} \;=\; \left\|\mathbf{v}_1 - \widehat{\mathbf{v}}_1\right\|^2.$$

Since $\|\mathbf{v}_1\|^2 = 1$ and $\mathbf{u}_1^T\mathbf{u}_2 = 0$, the resulting error is

$$e_{12} \;=\; 1 \;-\; \left(b_1^2 + b_2^2\right).$$

A.2 Experimental Grating Generation

The experimental stimuli used are sinusoidal gratings modulated by a Gaussian envelope. Each grating was mathematically defined as:

$$I(x,y) = c \cdot \cos\left(K\left(x\cos\theta + y\sin\theta\right) - \phi\right) \cdot \exp\left(-\frac{(x-x_0)^2 + (y-y_0)^2}{2\sigma^2}\right) \tag{46}$$

where c represents contrast, $K = 2\pi/\lambda$ is the spatial frequency with wavelength λ, θ is the orientation angle, and ϕ controls the phase shift. The exponential term applies a Gaussian envelope centered at (x_0, y_0) with spatial extent σ, and this gives us control over the aperture setting.